

METIMMOX-1 Value of Information Analysis

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

Table 1: Analysis scenario definitions. Higher WTP thresholds (€100,000 and €150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (€4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

Scenario Name	WTP Threshold	Nivolumab Cost
Base Case	€ 51,000	€13,923
Decreased Nivolumab Cost	€ 51,000	€ 4,641
WTP €100,000	€100,000	€13,923
WTP €100,000 + Decreased Nivolumab	€100,000	€ 4,641
WTP €150,000	€150,000	€13,923
WTP €150,000 + Decreased Nivolumab	€150,000	€ 4,641

2 Base Case Scenario

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €13,923

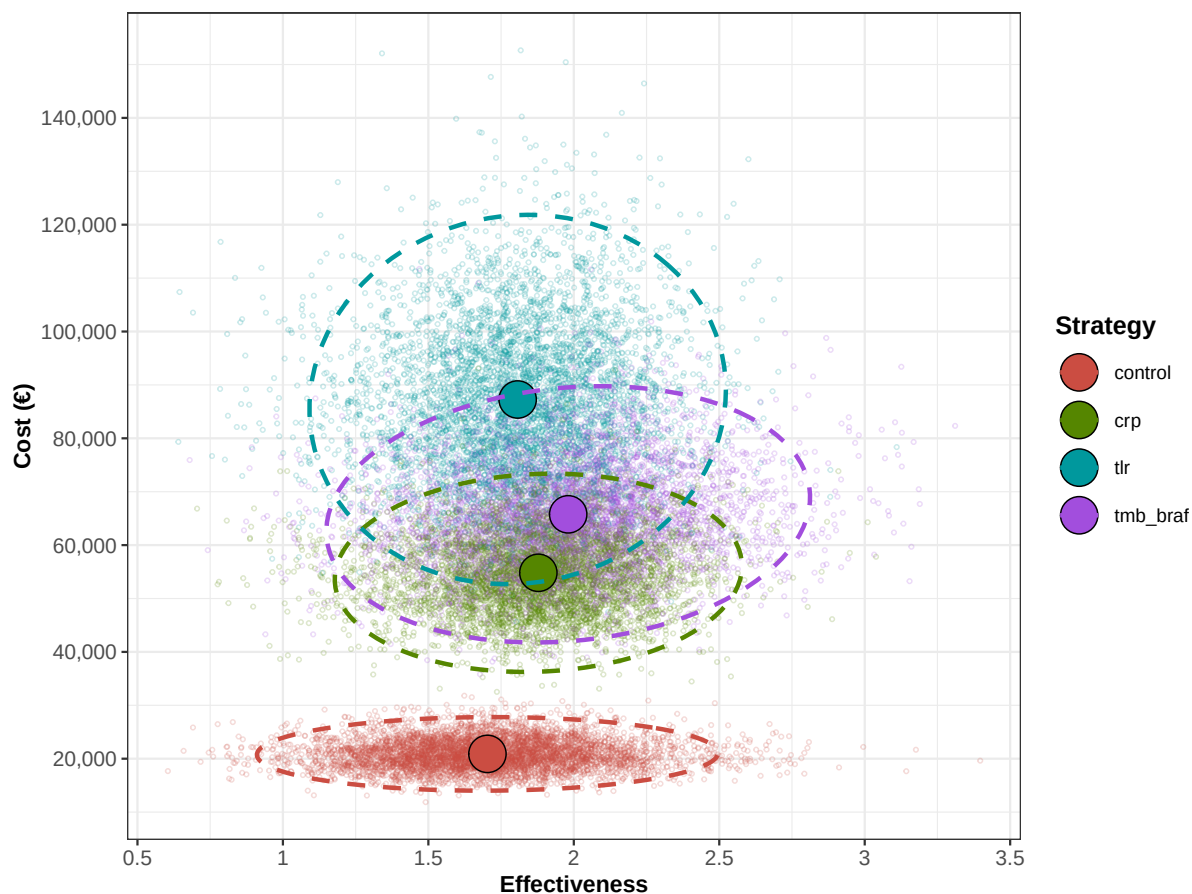


Figure 1: Cost-effectiveness scatter plot (base case)

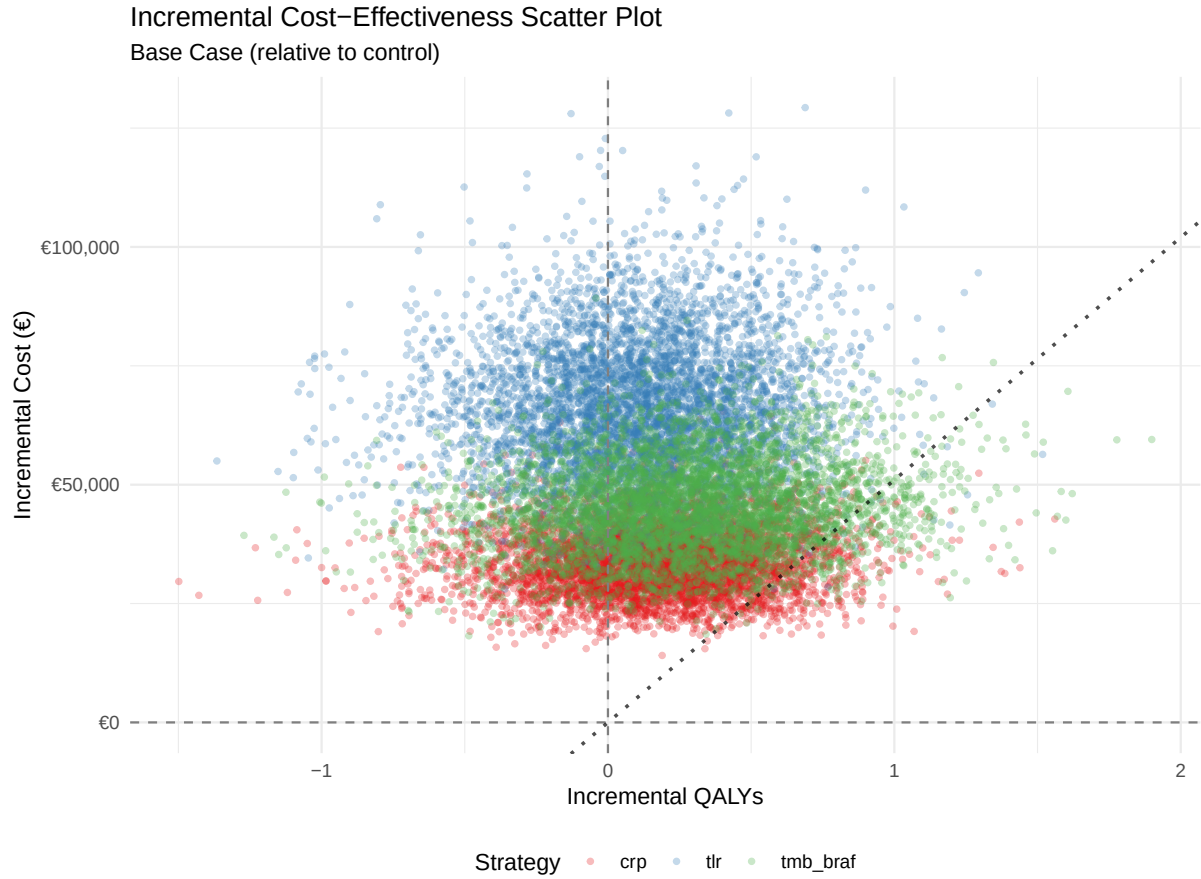


Figure 2: Incremental cost-effectiveness scatter plot relative to control (base case)

Table 2: Cost-effectiveness analysis summary from probabilistic sensitivity analysis (base case)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€20,901.54	€ 2,814.42	1.703	0.324	€65,954.41	€16,683.83
crp	€54,814.77	€ 7,585.36	1.878	0.286	€40,965.96	€15,980.55
tlr	€87,269.97	€14,129.49	1.807	0.292	€ 4,886.71	€20,014.10
tmb_braf	€65,764.24	€ 9,804.48	1.981	0.340	€35,252.49	€18,724.14

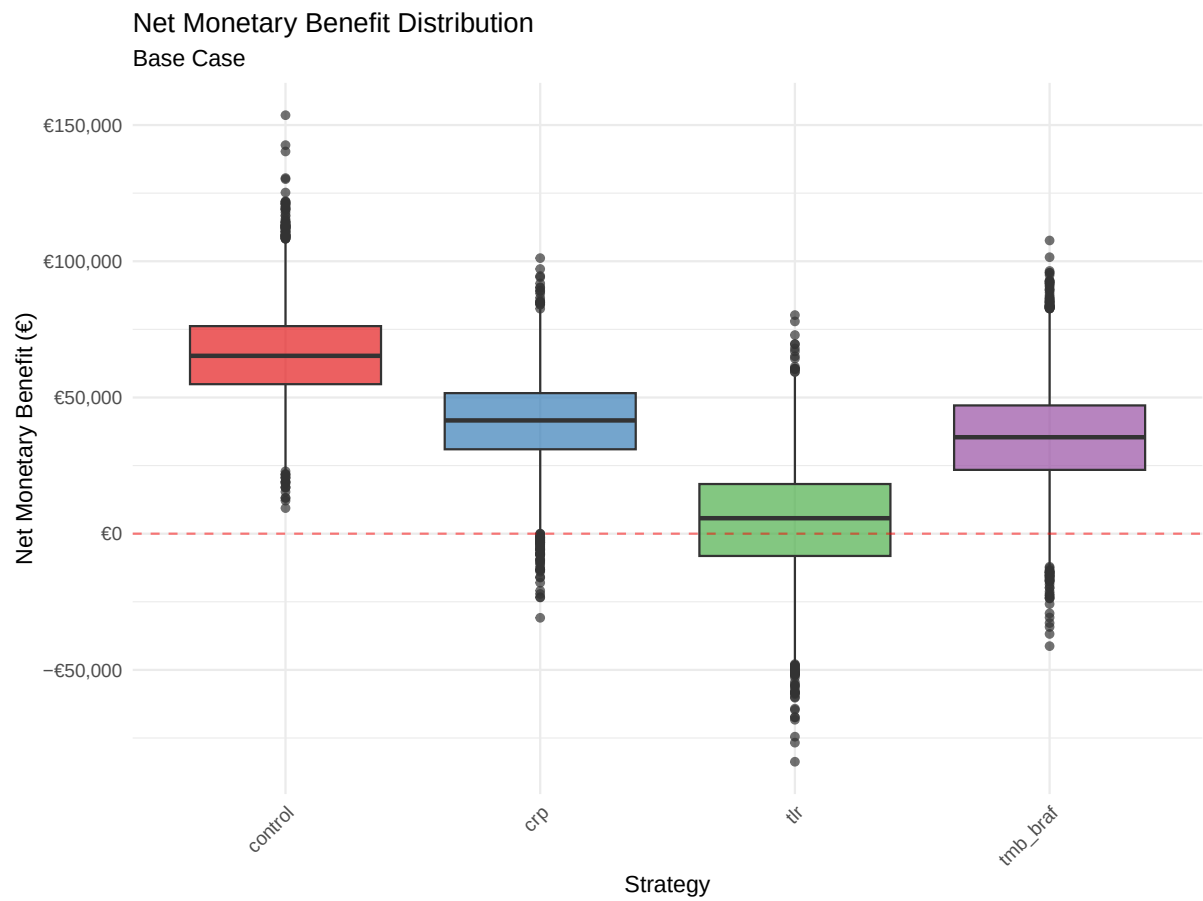


Figure 3: Net Monetary Benefit distribution (base case)

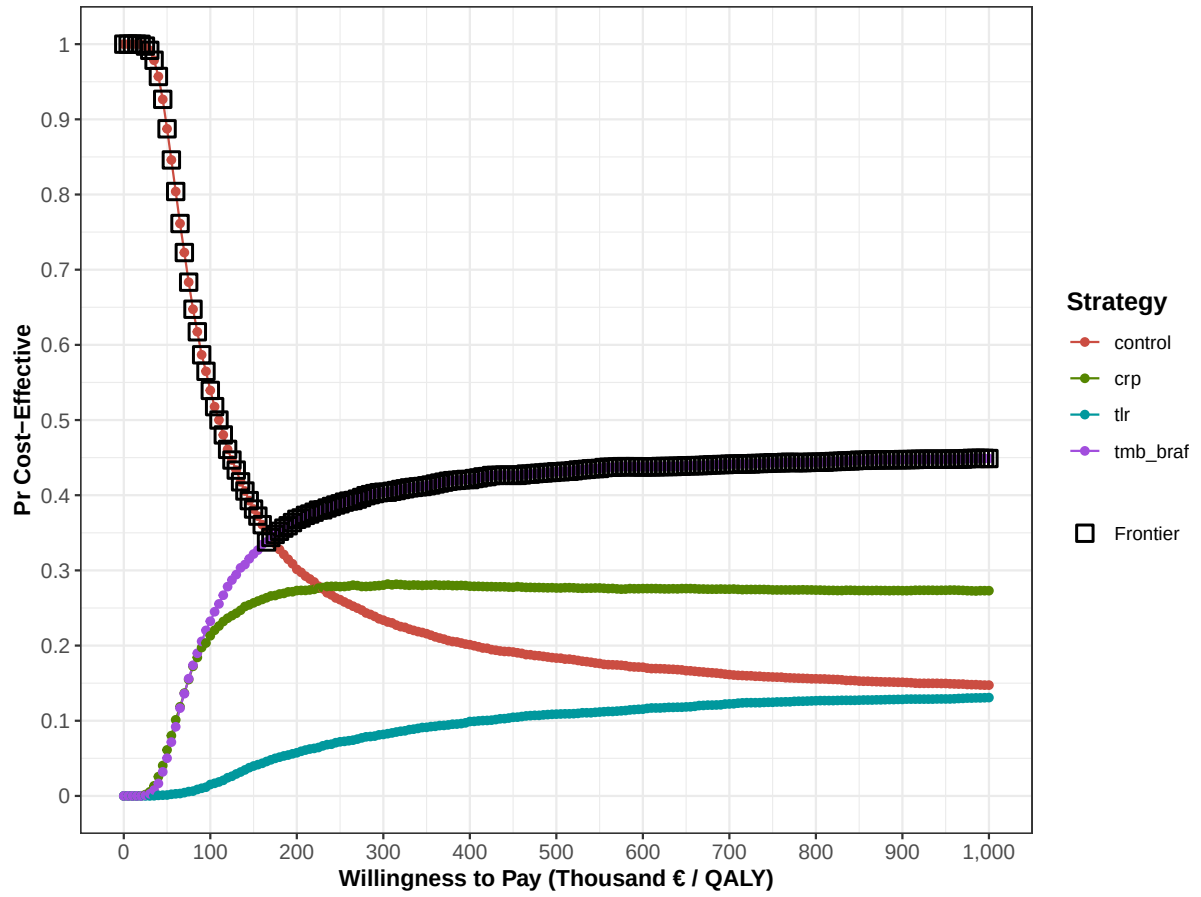


Figure 4: Cost-effectiveness acceptability curves (base case)

Total EVPI: €1064

The positive EVPI indicates decision uncertainty at the base case WTP threshold. This suggests that additional research to reduce parameter uncertainty could have value.

Table 3: Base case EVPPI results (ranked by importance)

Parameter	EVPI	% of EVPI
u_np	€967.84	91%
u_p	€899.14	84.5%
c_other_last	€76.71	7.2%
c_test_NGS	€63.26	5.9%
c_drug_nivo	€45.7	4.3%
c_test_CT	€20.63	1.9%
c_other_baseline	€5.72	0.5%
c_drug_FLOX	€0	0%
c_other_visit	€0	0%
c_other_follow	€0	0%

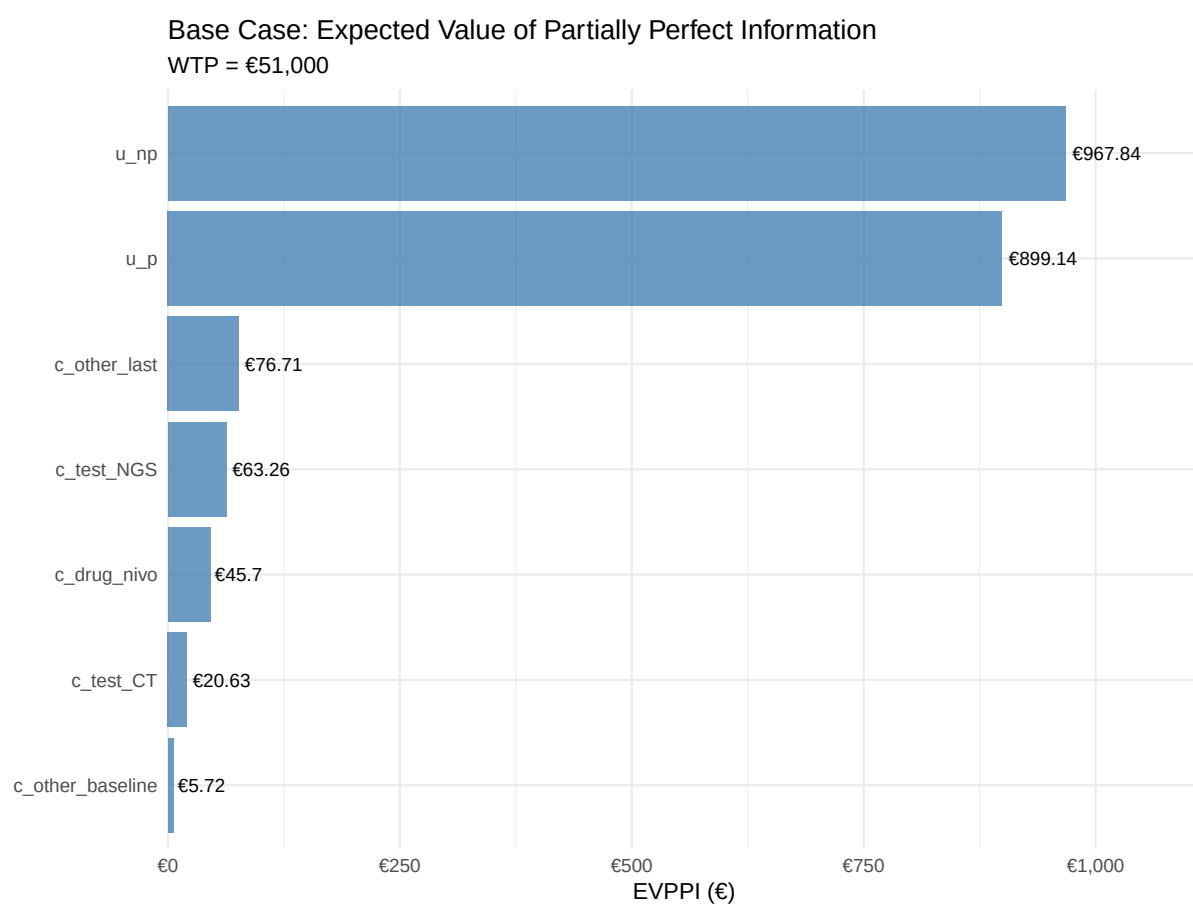


Figure 5: Base case EVPPPI by parameter

3 Scenario 1: Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €4,641 (80% discount)

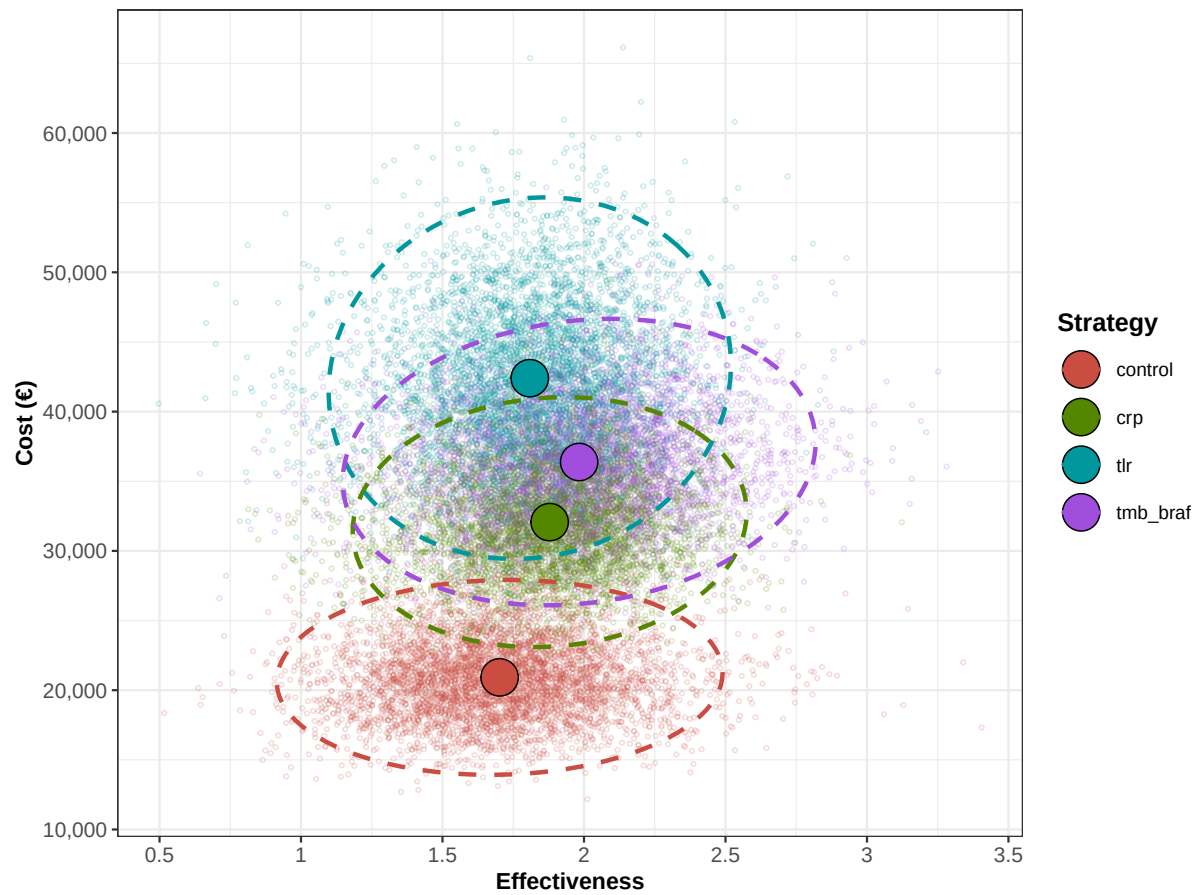


Figure 6: Cost-effectiveness scatter plot (Scenario 1)

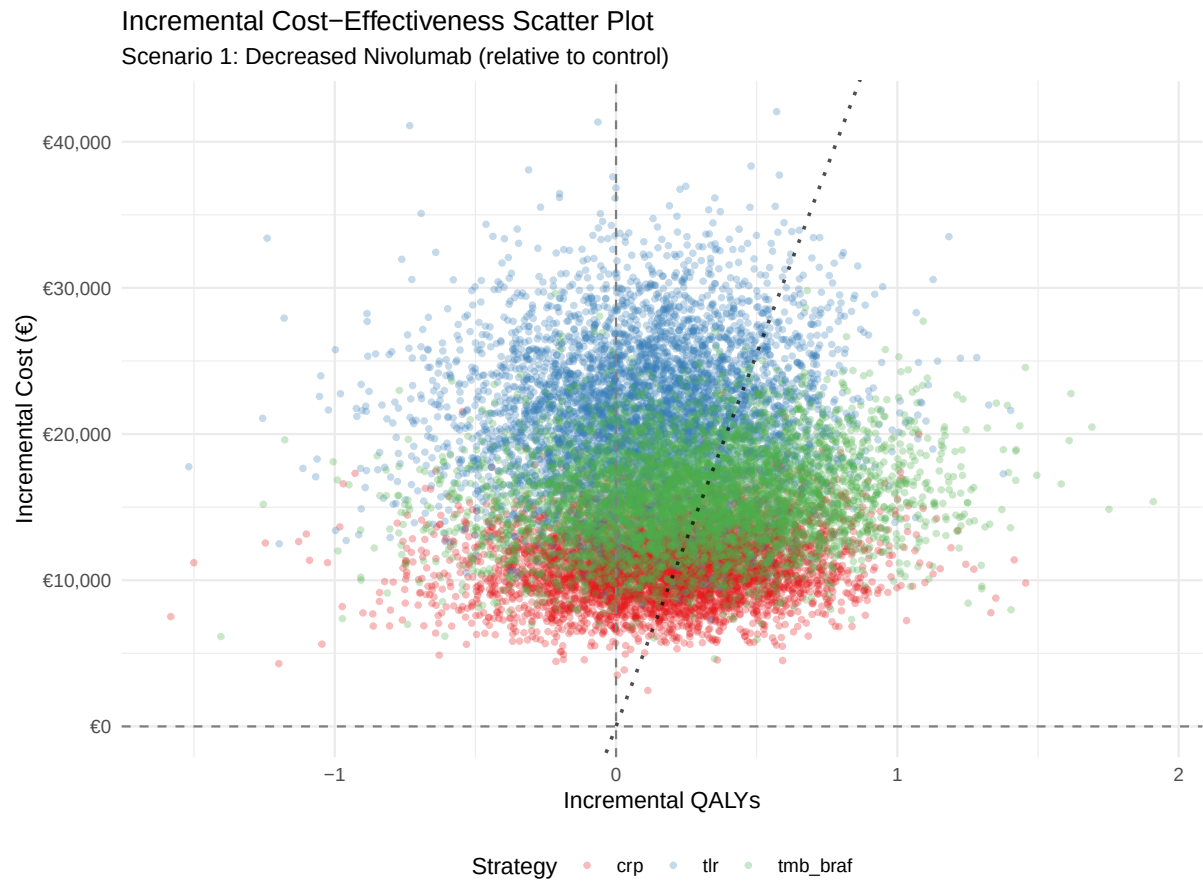


Figure 7: Incremental cost-effectiveness scatter plot relative to control (Scenario 1)

Table 4: Cost-effectiveness analysis summary (Scenario 1)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€20,920.28	€2,858.76	1.702	0.322	€65,876.27	€16,537.92
crp	€32,066.99	€3,665.47	1.879	0.284	€63,742.88	€14,684.21
tlr	€42,401.00	€5,302.43	1.808	0.290	€49,828.59	€15,346.51
tmb_braf	€36,379.51	€4,201.35	1.983	0.341	€64,745.38	€17,346.16

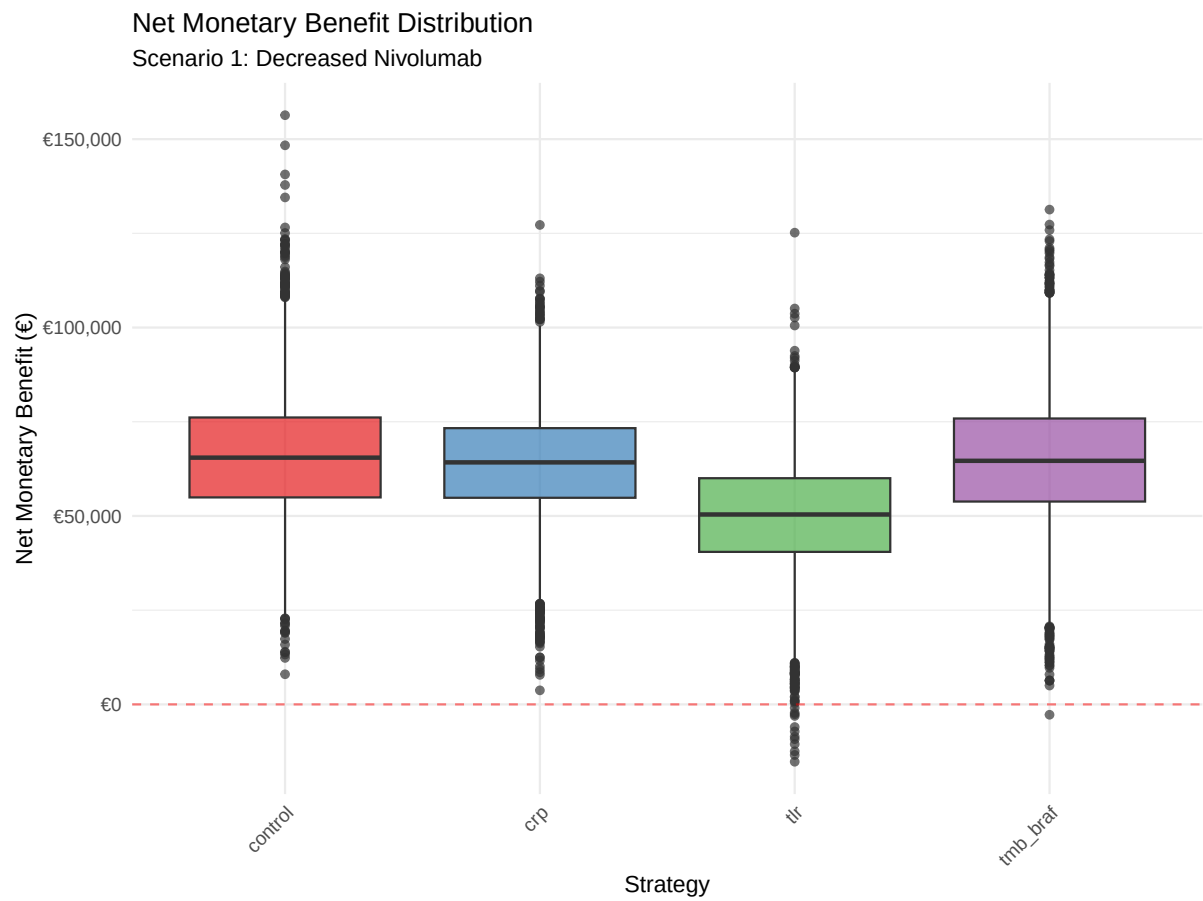


Figure 8: Net Monetary Benefit distribution (Scenario 1)

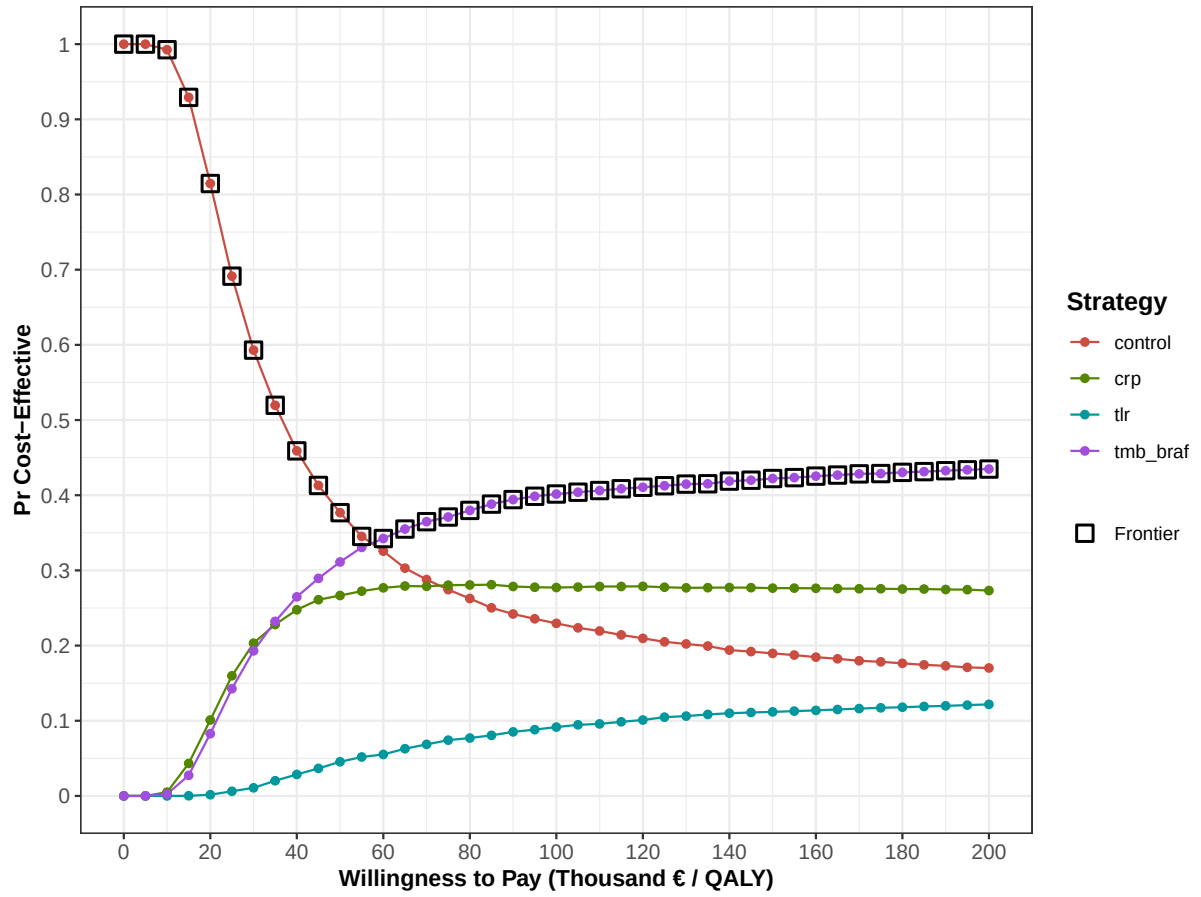


Figure 9: Cost-effectiveness acceptability curves (Scenario 1)

Total EVPI: €9887.23

Reducing nivolumab costs creates decision uncertainty, making EVPPI analysis meaningful.

Table 5: EVPPI results: Decreased nivolumab cost

Parameter	EVPPI	% of EVPI
u_np	€1127.63	11.4%
u_p	€923.44	9.3%
c_drug_nivo	€885.25	9%
c_drug_FLOX	€65.67	0.7%
c_other_visit	€47.41	0.5%
c_other_last	€14.16	0.1%
c_test_NGS	€0	0%
c_test_CT	€0	0%
c_other_baseline	€0	0%
c_other_follow	€0	0%

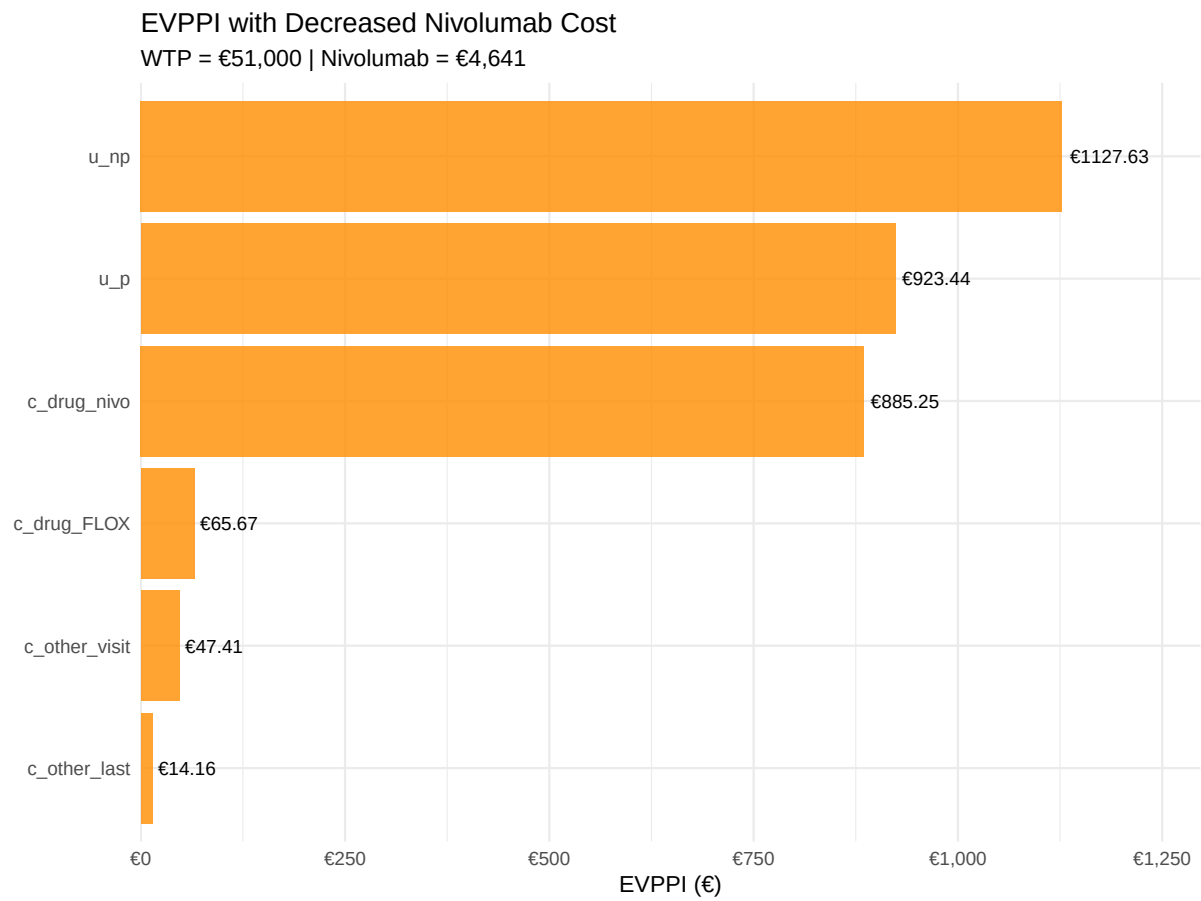


Figure 10: EVPI with decreased nivolumab cost

4 Scenario 2: Higher WTP Threshold (€100,000)

Willingness-to-Pay Threshold: €1e+05

Nivolumab Cost: €13,923 (standard)

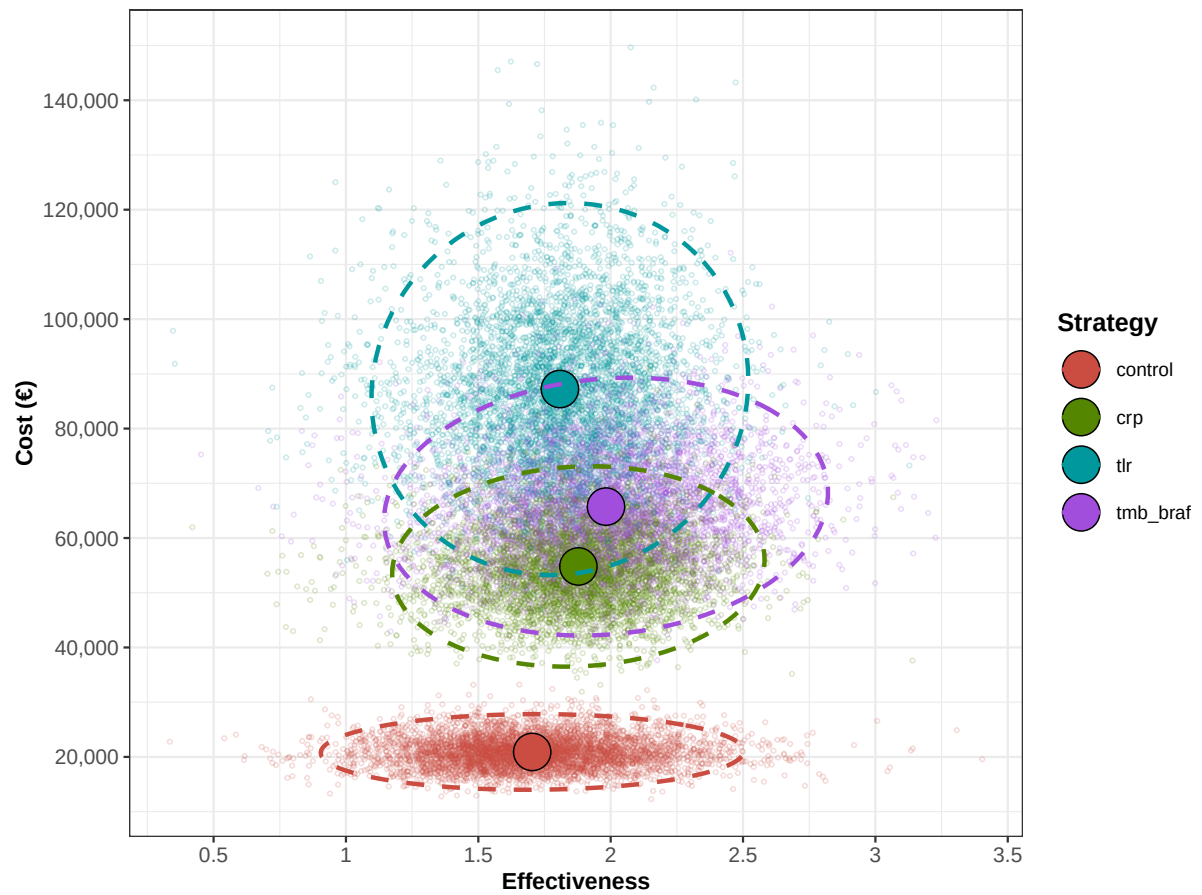


Figure 11: Cost-effectiveness scatter plot (Scenario 2)

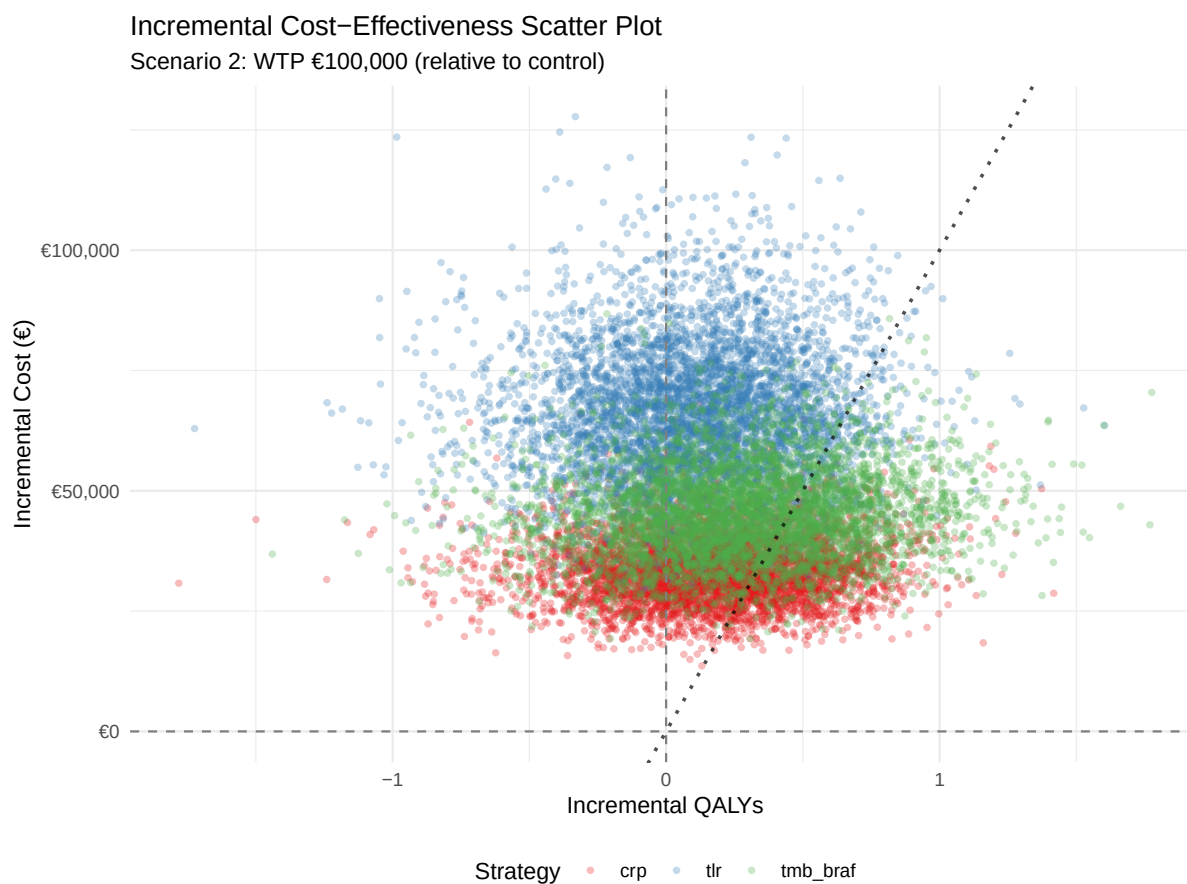


Figure 12: Incremental cost-effectiveness scatter plot relative to control (Scenario 2)

Table 6: Cost-effectiveness analysis summary (Scenario 2)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€20,910.34	€ 2,823.12	1.704	0.326	€149,450.66	€32,696.61
crp	€54,807.46	€ 7,482.21	1.878	0.287	€132,993.31	€29,200.50
tlr	€87,230.57	€13,889.31	1.808	0.291	€ 93,588.52	€31,607.15
tmb_braf	€65,745.14	€ 9,626.96	1.982	0.343	€132,490.71	€34,562.19

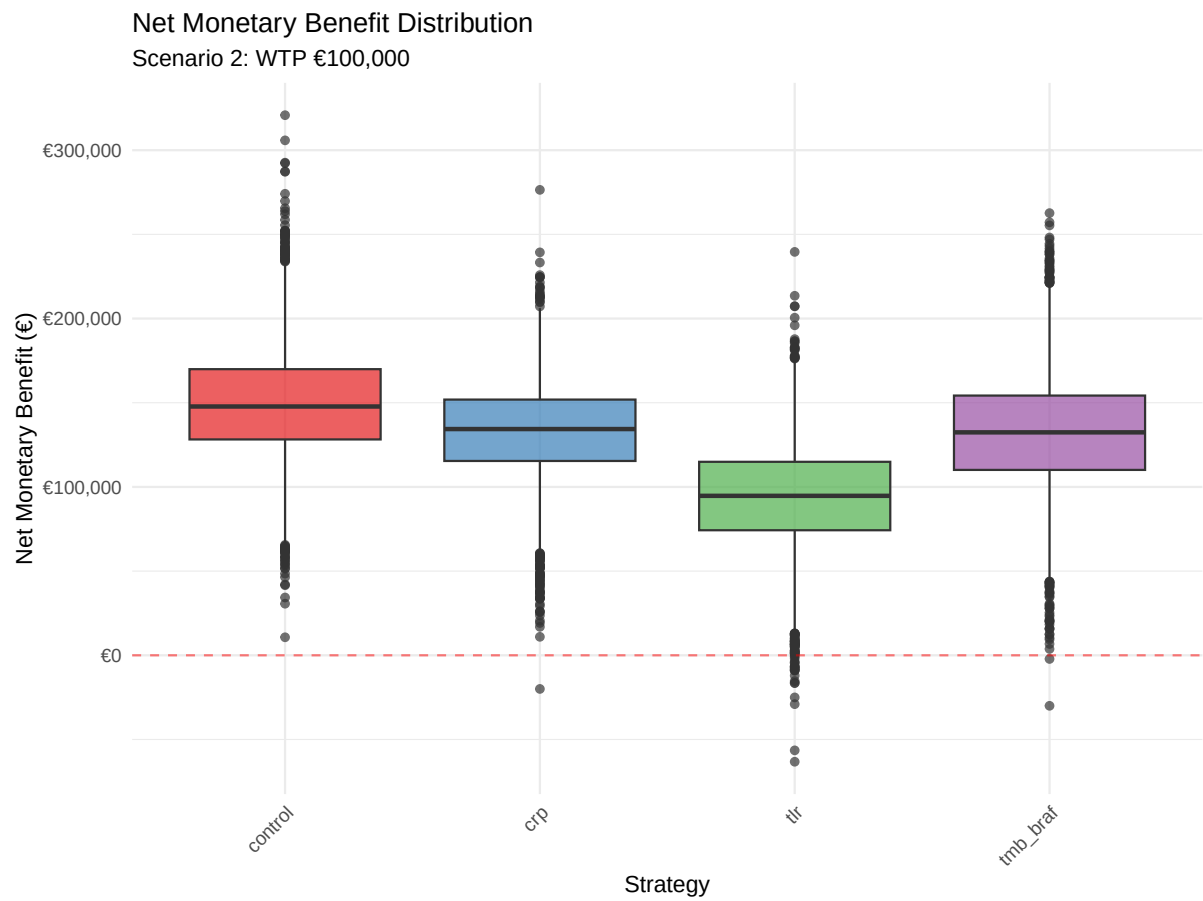


Figure 13: Net Monetary Benefit distribution (Scenario 2)

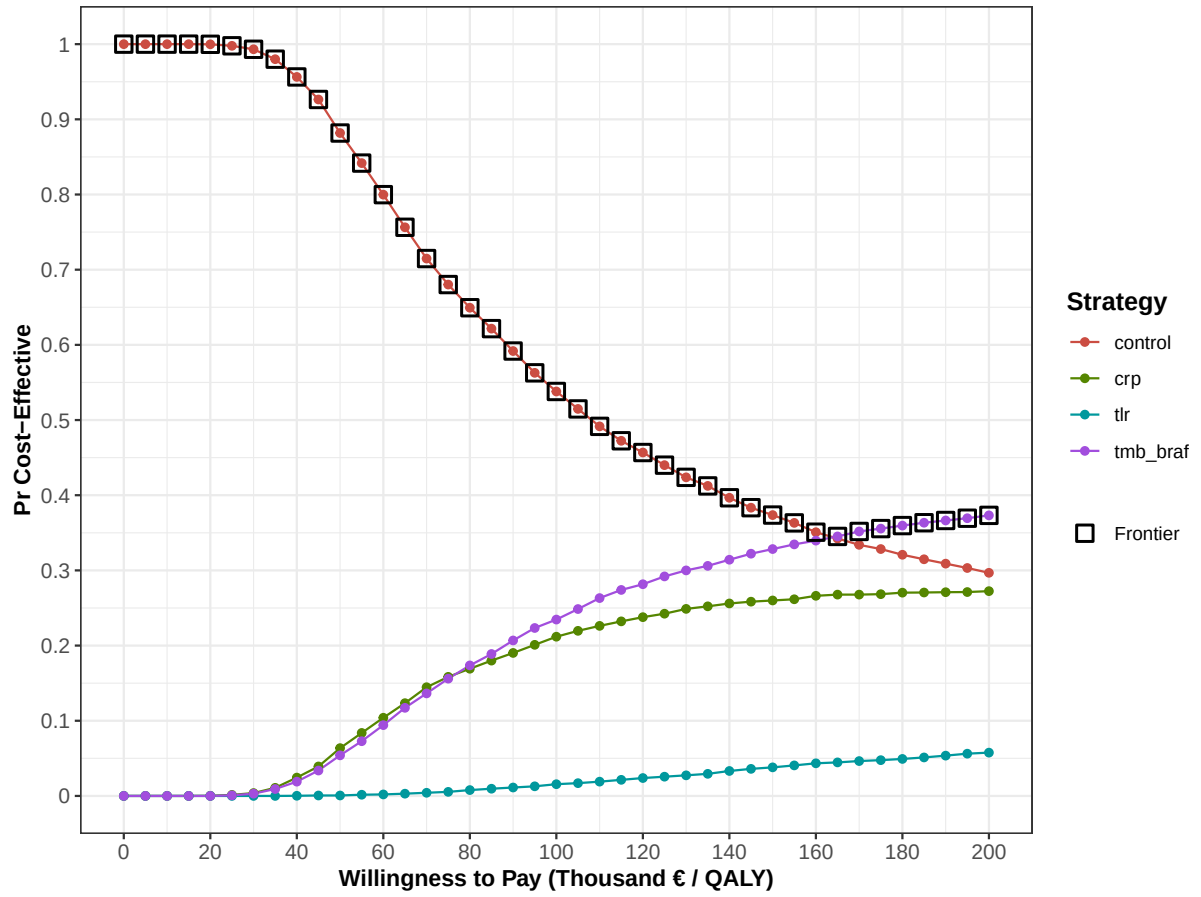


Figure 14: Cost-effectiveness acceptability curves (Scenario 2)

Total EVPI: $\text{€}1.198463 \times 10^4$

Table 7: EVPPI results: WTP €100,000

Parameter	EVPPI	% of EVPI
u_np	€1865.01	15.6%
u_p	€1800.89	15%
c_other_last	€237.04	2%
c_drug_nivo	€93.27	0.8%
c_other_baseline	€91.51	0.8%
c_test_CT	€49.71	0.4%
c_other_follow	€36.94	0.3%
c_drug_FLOX	€33.74	0.3%
c_test_NGS	€0	0%
c_other_visit	€0	0%

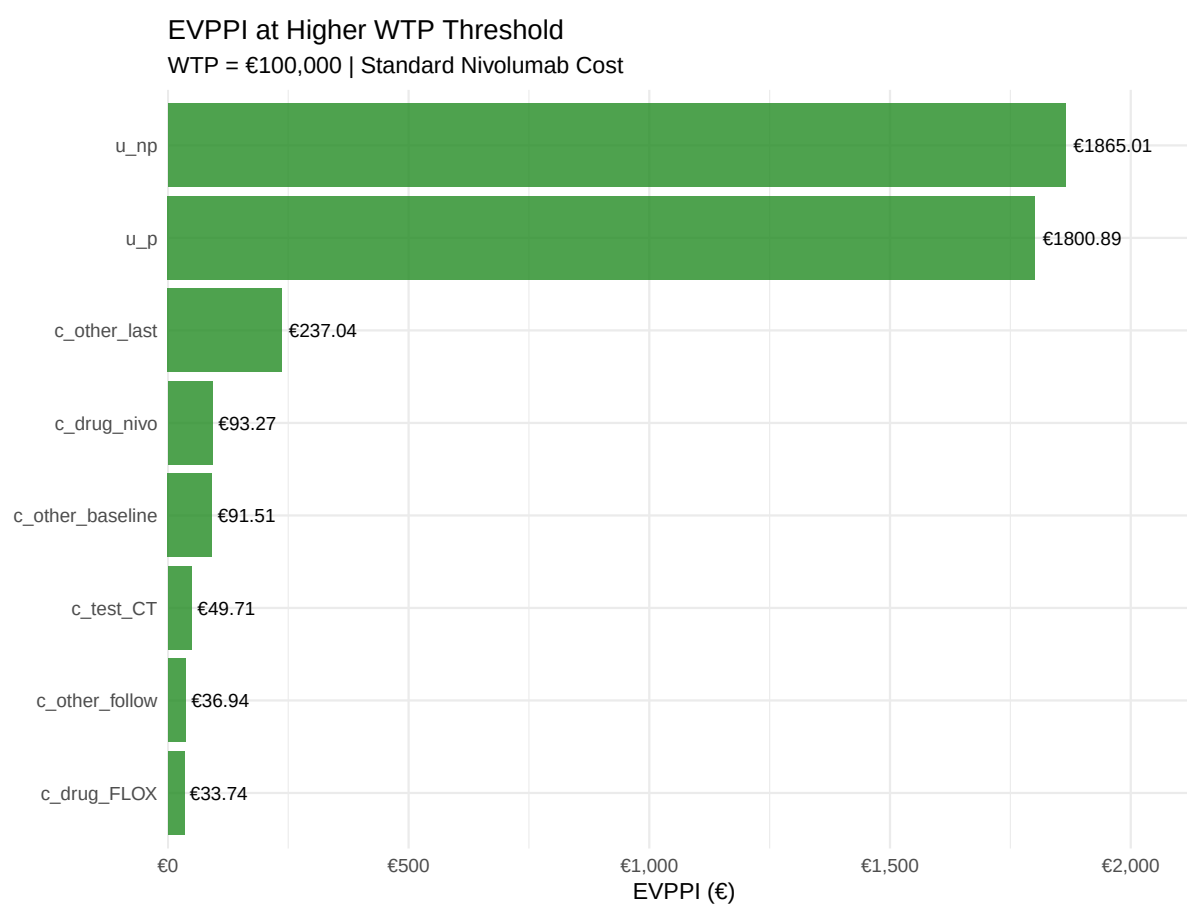


Figure 15: EVPPI at WTP €100,000

5 Scenario 3: Higher WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €1e+05

Nivolumab Cost: €4,641 (80% discount)

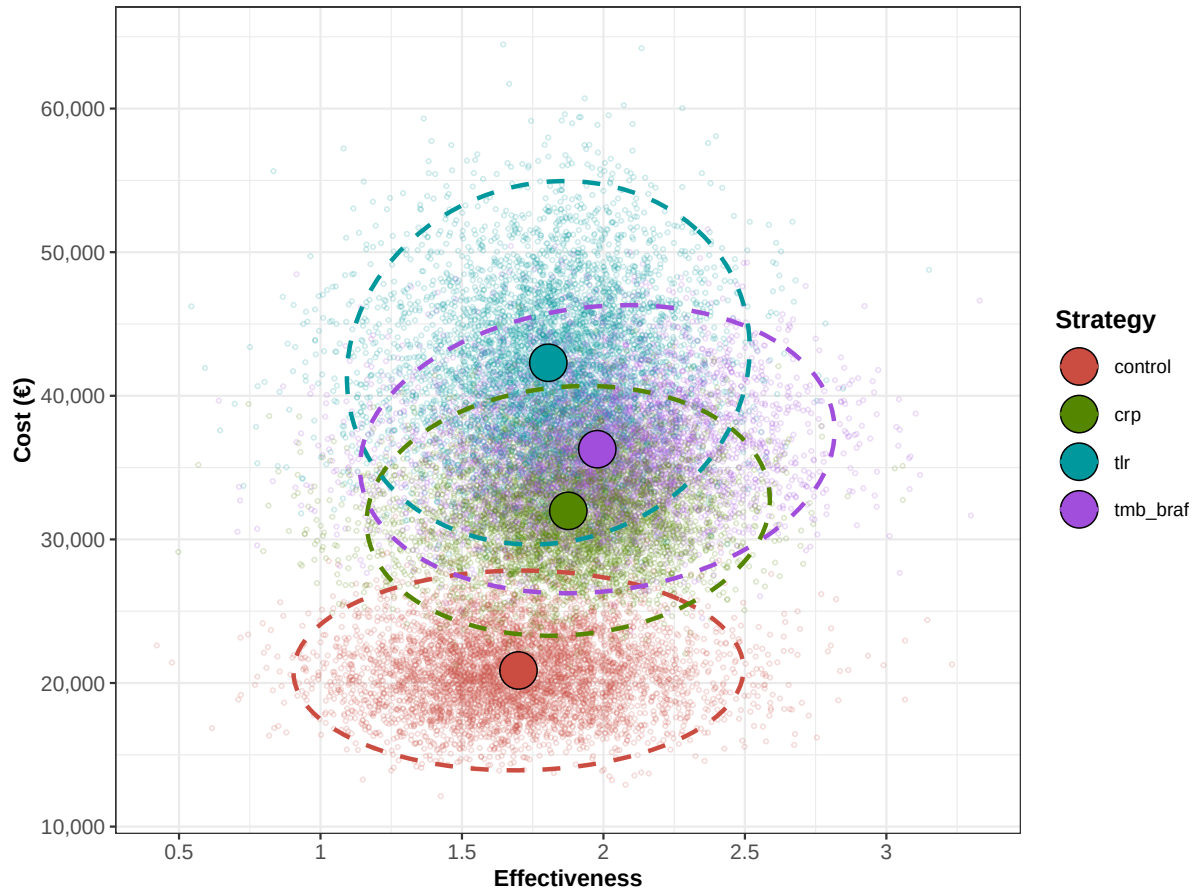


Figure 16: Cost-effectiveness scatter plot (Scenario 3)

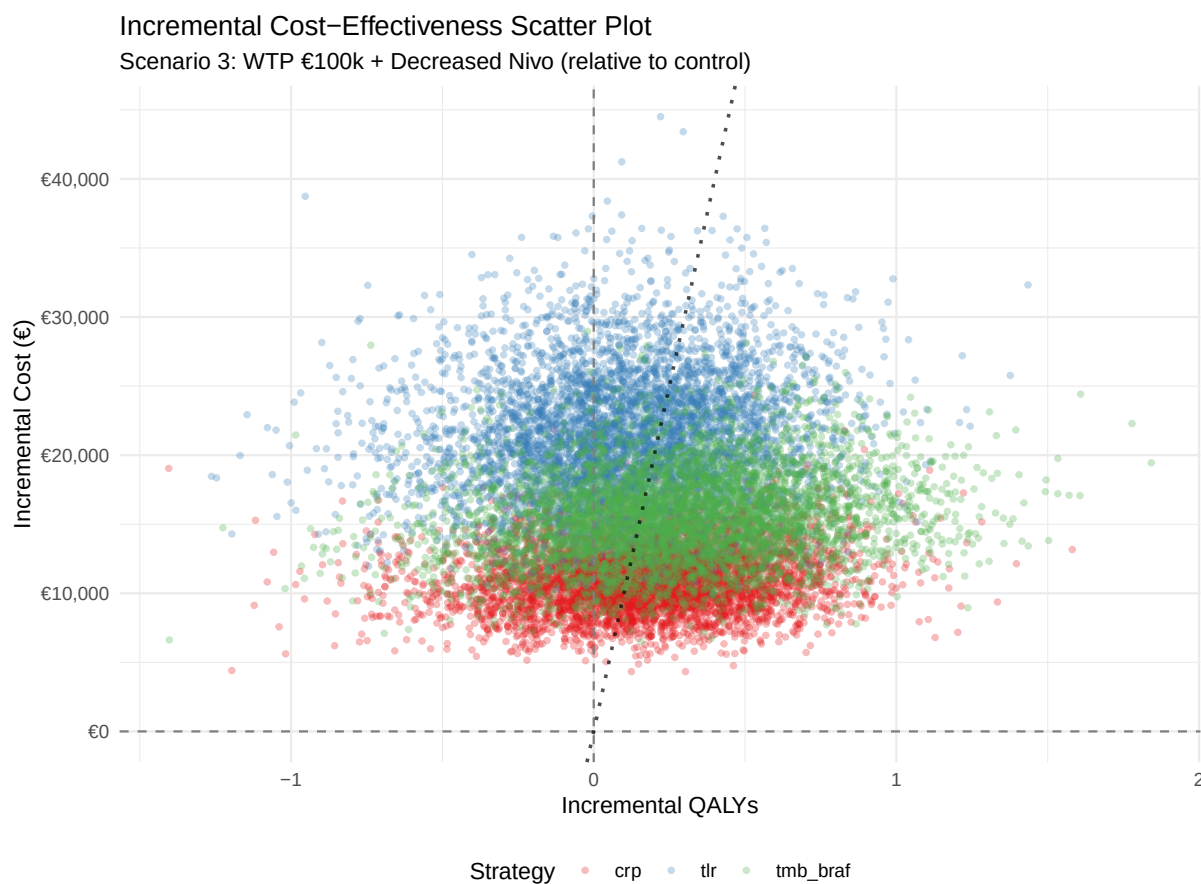


Figure 17: Incremental cost-effectiveness scatter plot relative to control (Scenario 3)

Table 8: Cost-effectiveness analysis summary (Scenario 3)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€20,873.97	€2,842.39	1.700	0.325	€149,107.2	€32,576.91
crp	€31,986.49	€3,553.56	1.876	0.291	€155,594.4	€29,011.84
tlr	€42,298.49	€5,170.36	1.805	0.291	€138,211.0	€29,125.55
tmb_braf	€36,280.40	€4,099.22	1.978	0.343	€161,548.0	€33,990.74

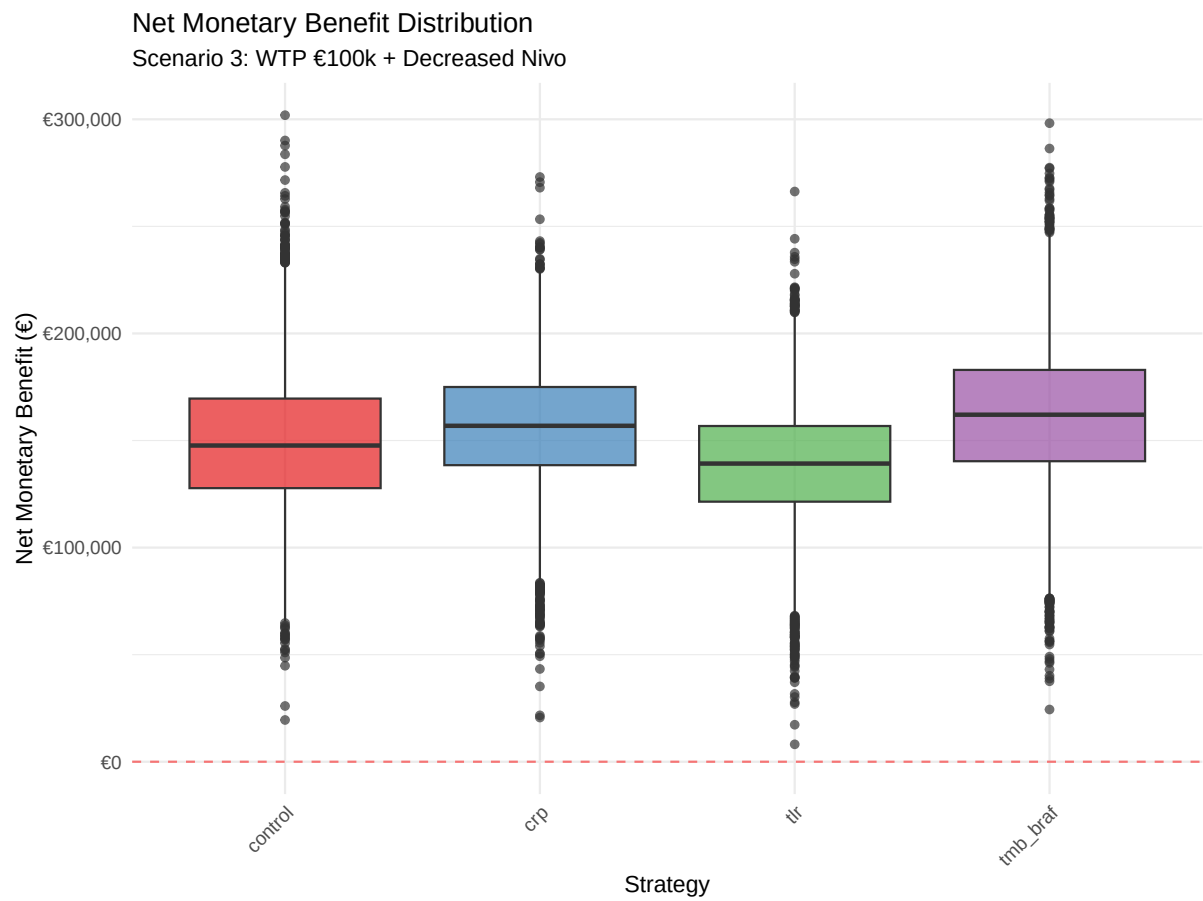


Figure 18: Net Monetary Benefit distribution (Scenario 3)

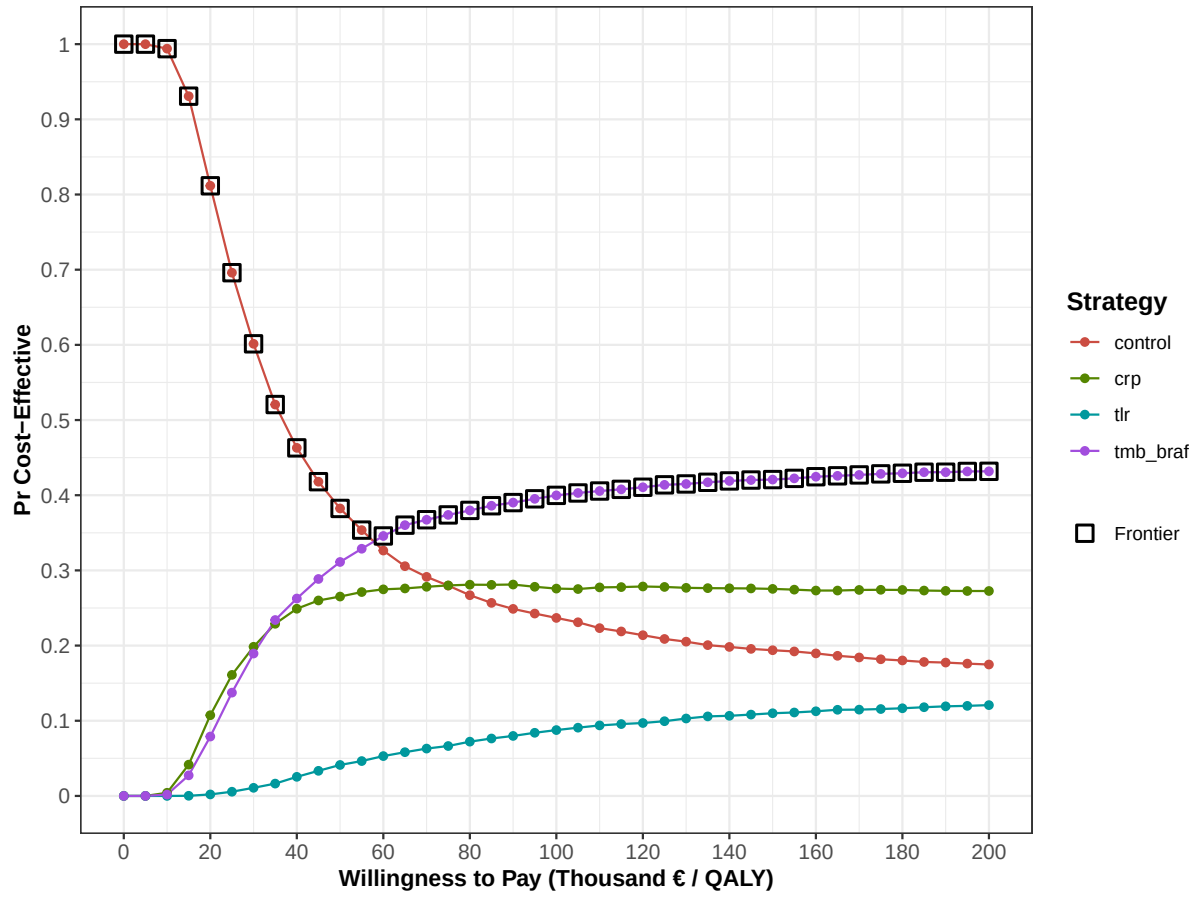


Figure 19: Cost-effectiveness acceptability curves (Scenario 3)

Total EVPI: $\text{€}1.657382 \times 10^4$

Table 9: EVPPI results: WTP $\text{€}100,000$ + decreased nivolumab

Parameter	EVPPI	% of EVPI
u_np	€2568.1	15.5%
u_p	€1383.44	8.3%
c_other_visit	€211.07	1.3%
c_drug_nivo	€131.08	0.8%
c_drug_FLOX	€119.23	0.7%
c_other_last	€66.68	0.4%
c_test_NGS	€53.22	0.3%
c_other_baseline	€16.39	0.1%
c_test_CT	€0.79	0%
c_other_follow	€0	0%

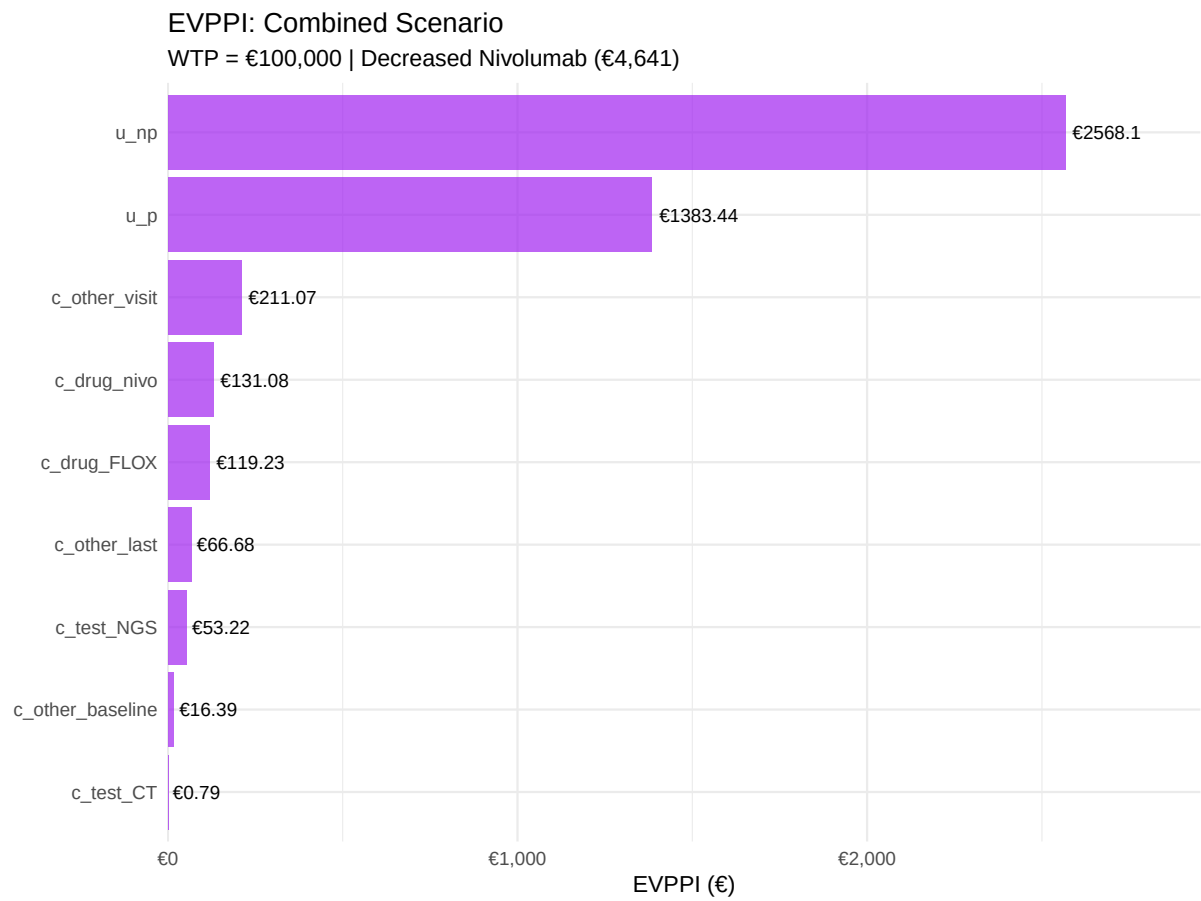


Figure 20: EVPPi: WTP €100,000 with decreased nivolumab

6 Scenario 4: Highest WTP Threshold (€150,000)

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: €13,923 (standard)

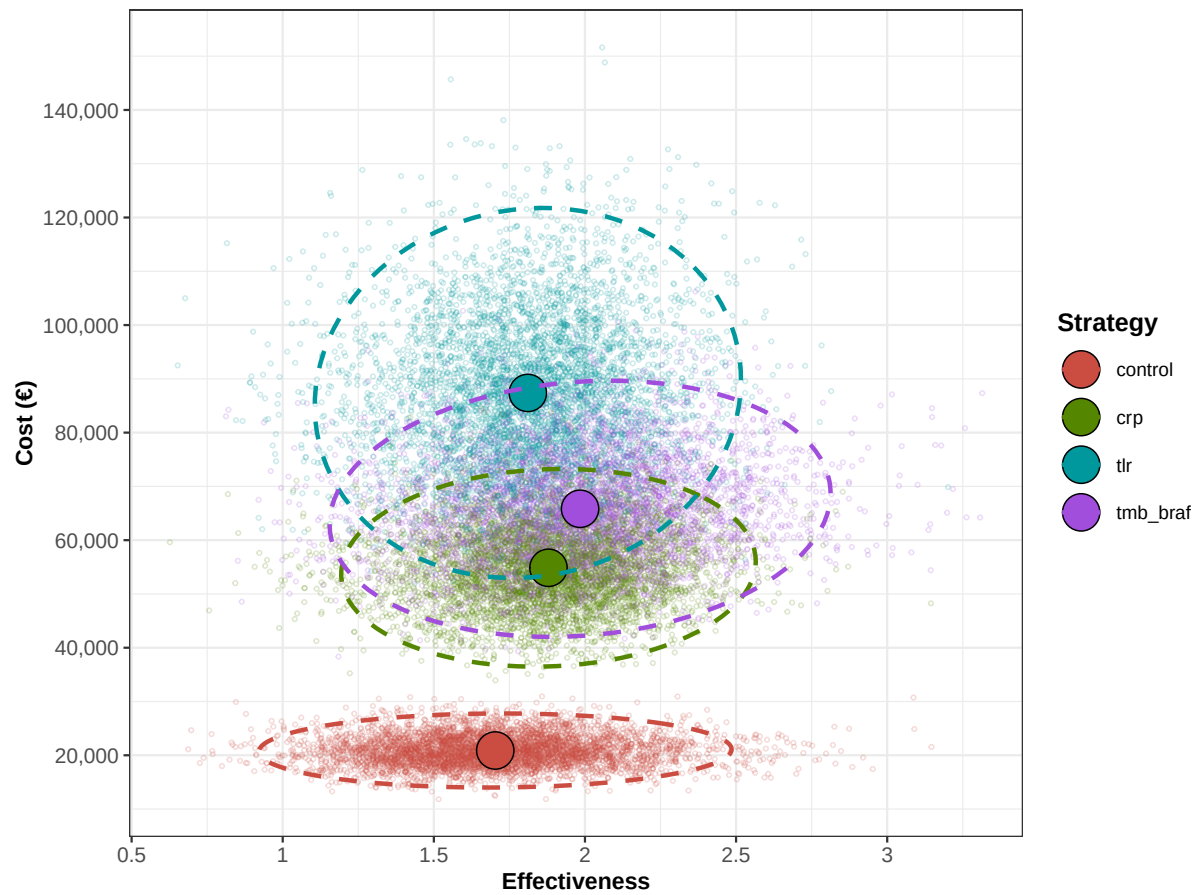


Figure 21: Cost-effectiveness scatter plot (Scenario 4)

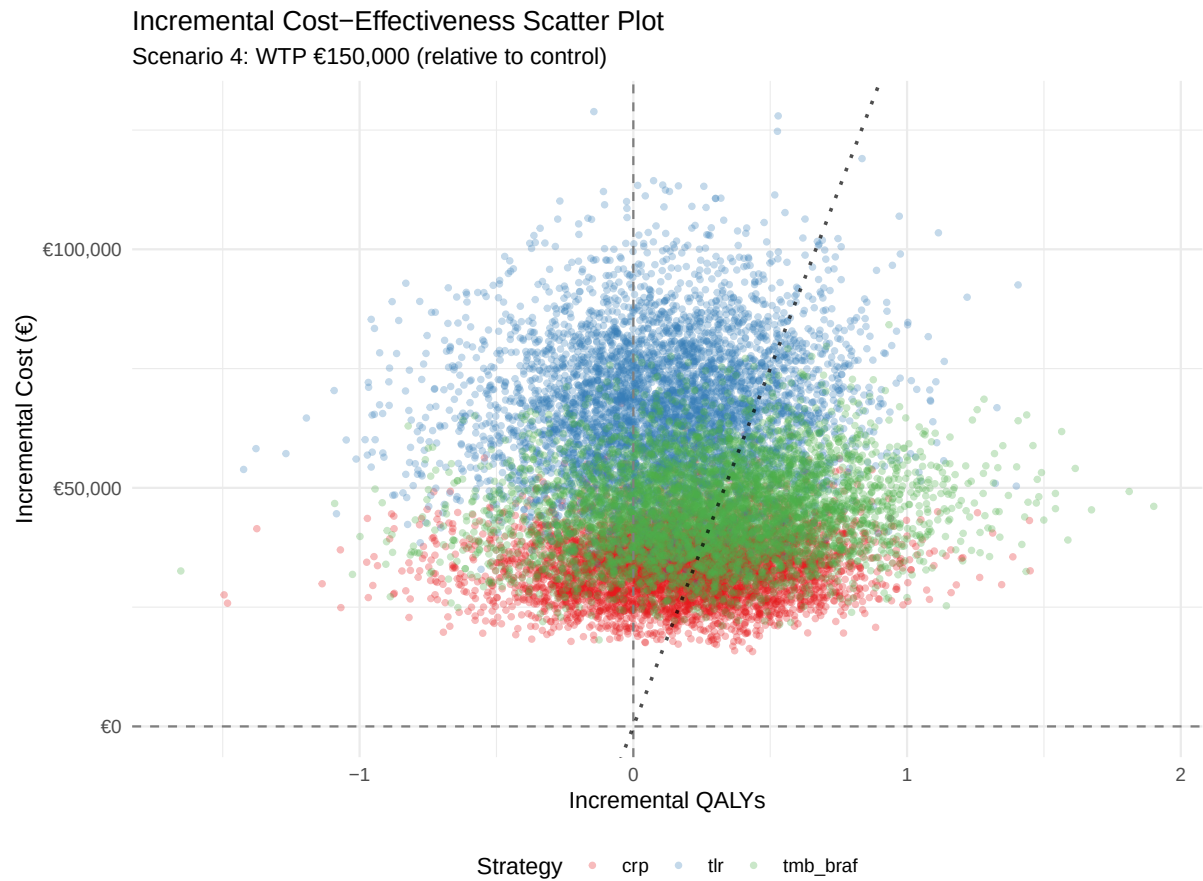


Figure 22: Incremental cost-effectiveness scatter plot relative to control (Scenario 4)

Table 10: Cost-effectiveness analysis summary (Scenario 4)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€20,886.25	€ 2,812.59	1.703	0.320	€234,587.8	€47,924.87
crp	€54,863.01	€ 7,510.78	1.879	0.280	€227,026.8	€42,216.01
tlr	€87,379.74	€14,056.50	1.811	0.288	€184,282.7	€44,495.40
tmb_braf	€65,837.32	€ 9,731.95	1.984	0.339	€231,734.0	€50,552.47

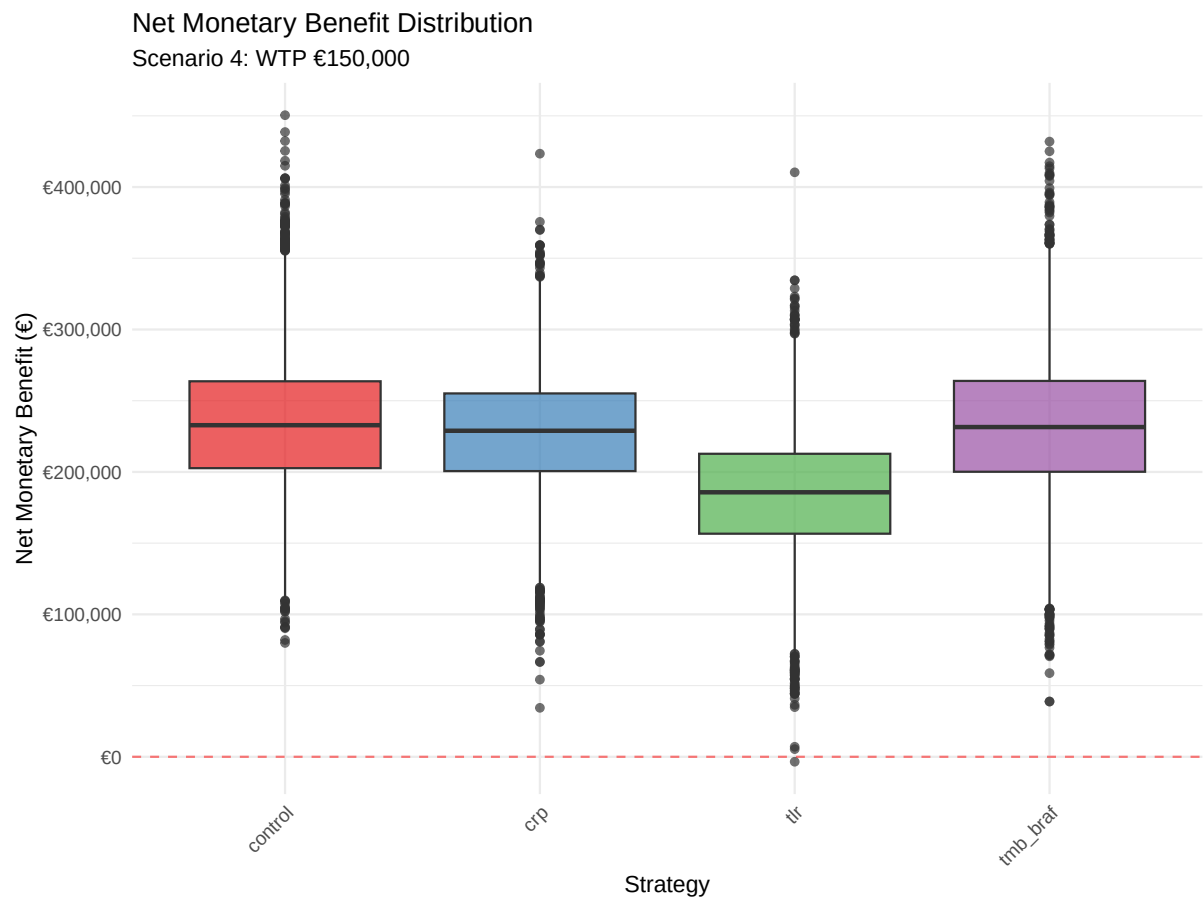


Figure 23: Net Monetary Benefit distribution (Scenario 4)

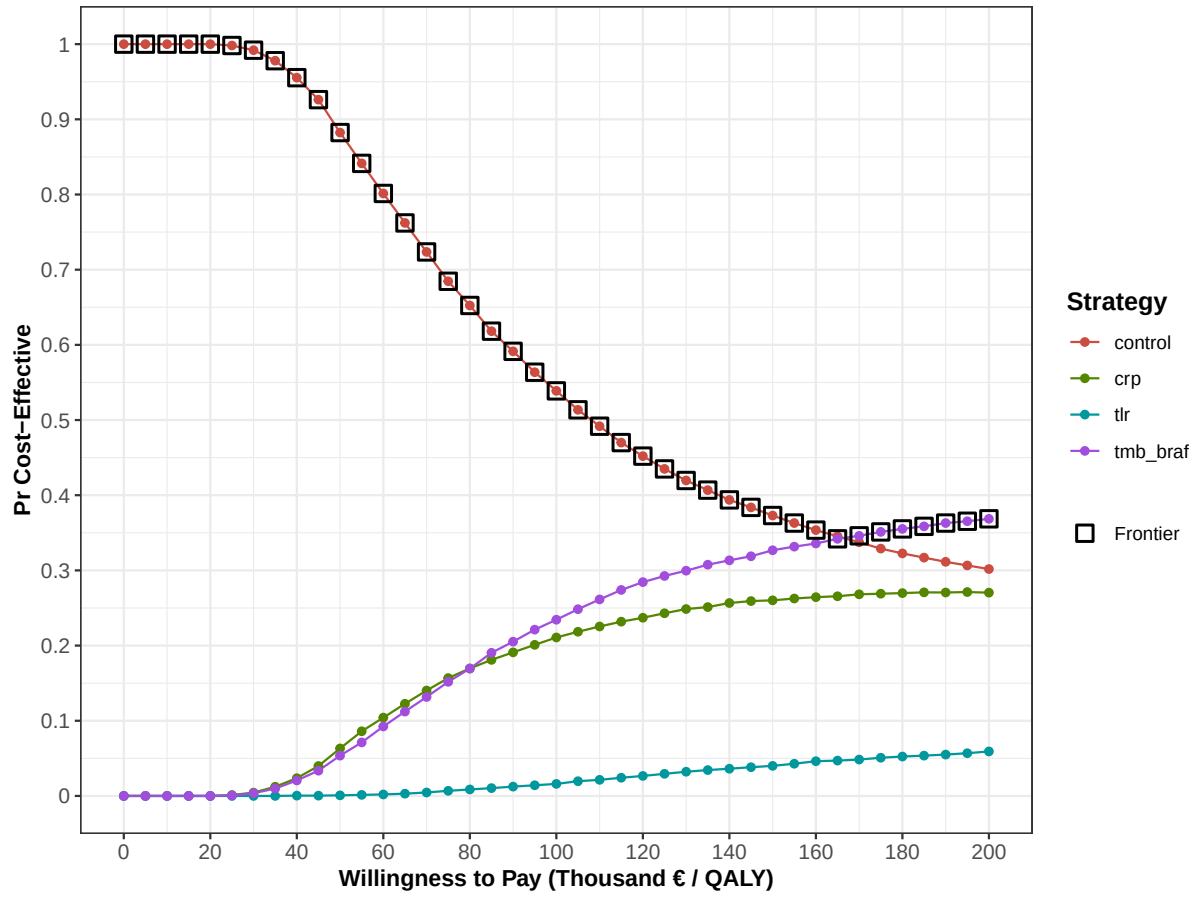


Figure 24: Cost-effectiveness acceptability curves (Scenario 4)

Total EVPI: $\text{€}2.91697 \times 10^4$

Table 11: EVPPI results: WTP €150,000

Parameter	EVPPI	% of EVPI
u_np	€3120.65	10.7%
u_p	€2322.3	8%
c_drug_nivo	€1839.28	6.3%
c_test_CT	€268.43	0.9%
c_other_follow	€124.3	0.4%
c_test_NGS	€94.04	0.3%
c_other_last	€69.88	0.2%
c_drug_FLOX	€24.45	0.1%
c_other_visit	€0	0%
c_other_baseline	€0	0%

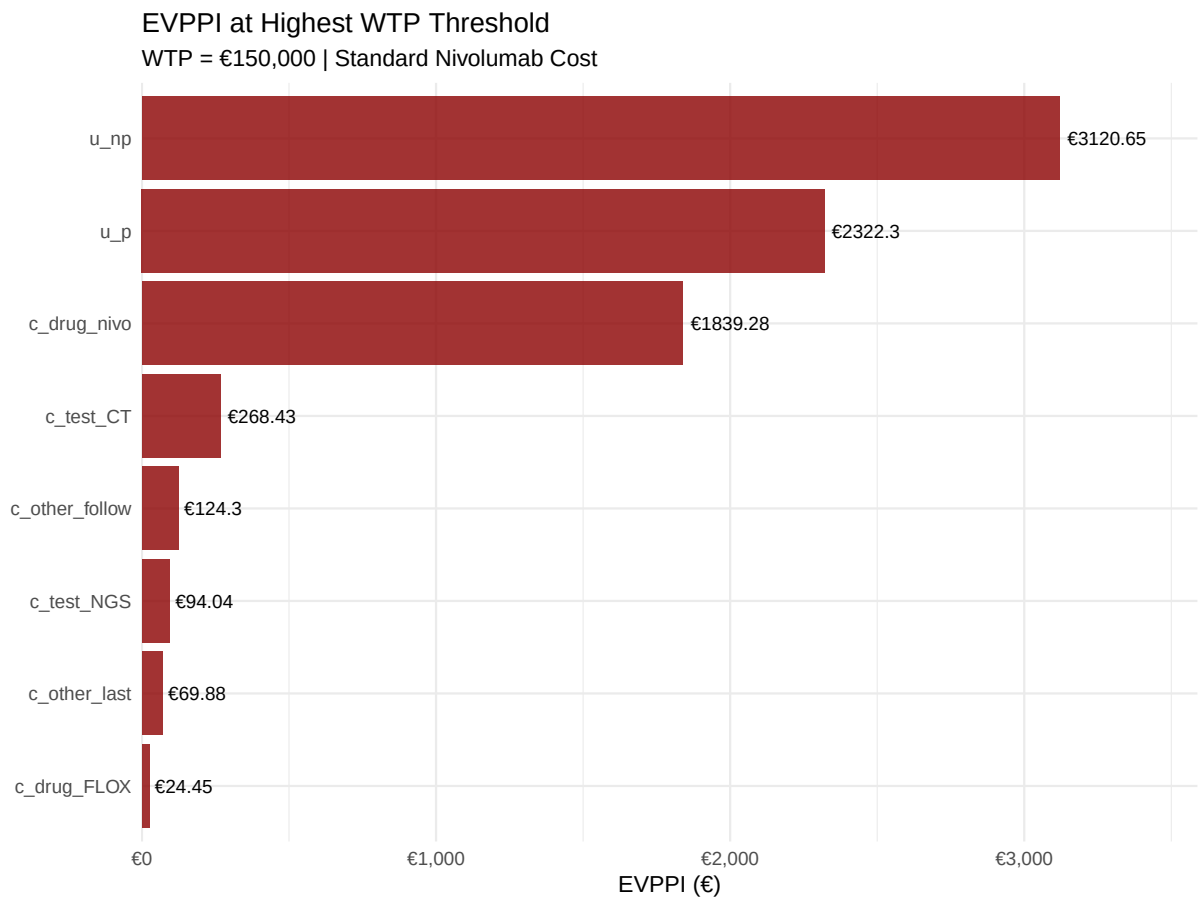


Figure 25: EVPPPI at WTP €150,000

7 Scenario 5: Highest WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: €4,641 (80% discount)

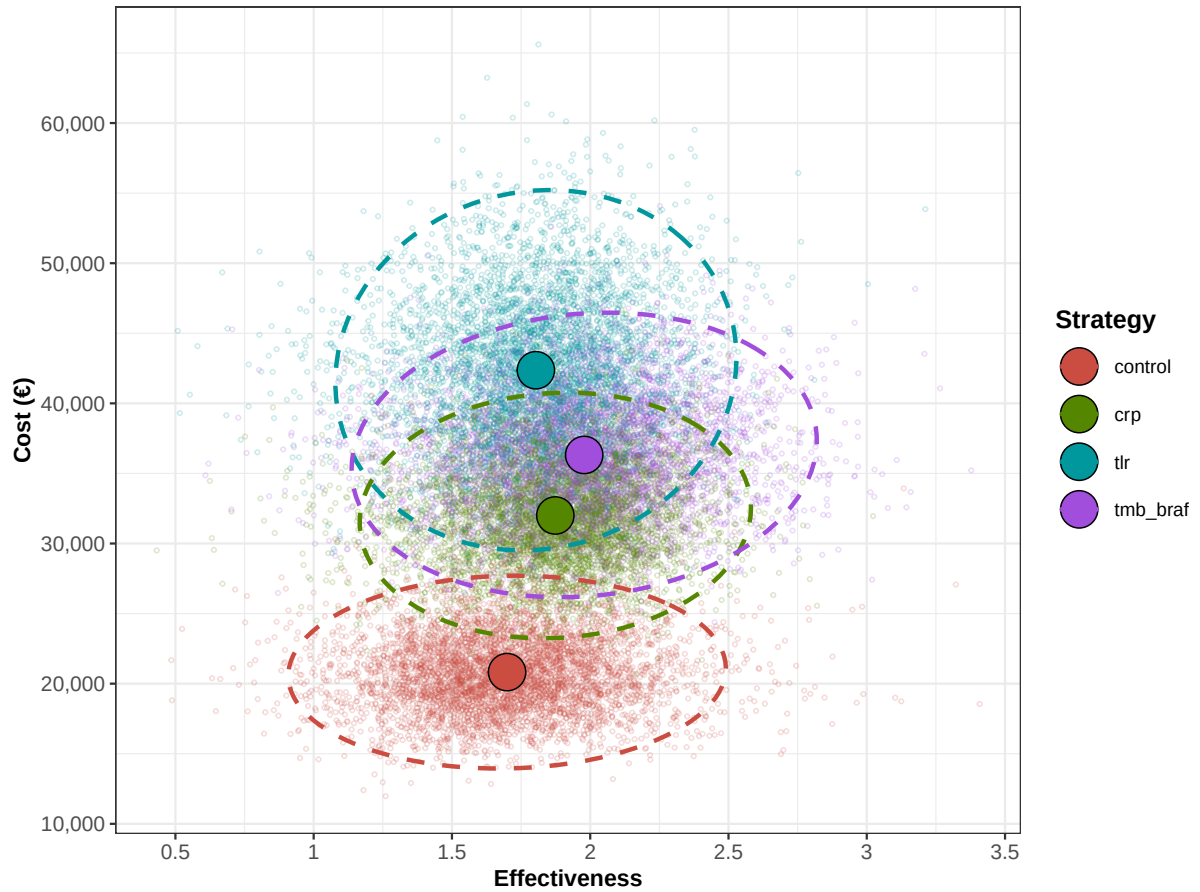


Figure 26: Cost-effectiveness scatter plot (Scenario 5)

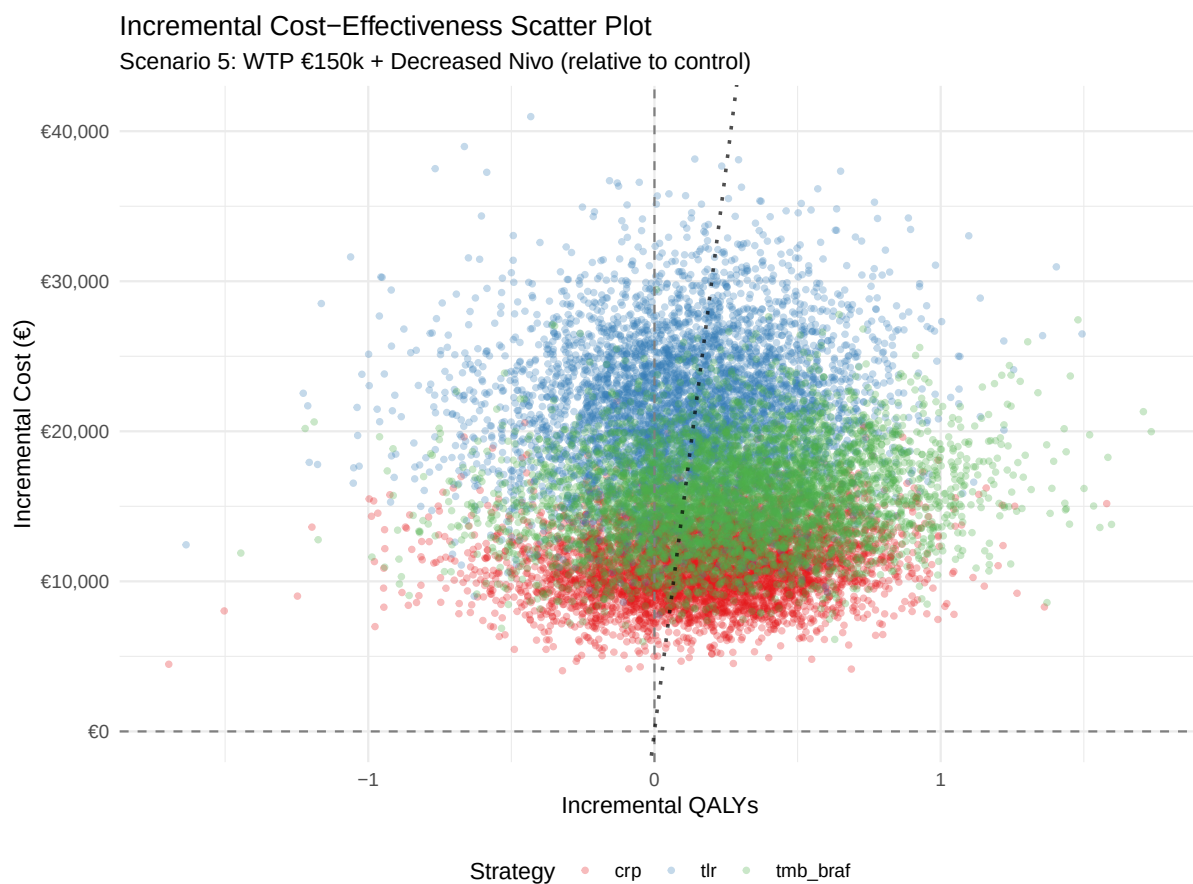


Figure 27: Incremental cost-effectiveness scatter plot relative to control (Scenario 5)

Table 12: Cost-effectiveness analysis summary (Scenario 5)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€20,824.30	€2,810.63	1.699	0.323	€234,077.4	€48,458.52
crp	€32,000.81	€3,580.51	1.874	0.289	€249,065.6	€43,288.26
tlr	€42,365.08	€5,251.53	1.804	0.296	€228,164.6	€44,389.45
tmb_braf	€36,318.96	€4,148.76	1.978	0.344	€260,414.0	€51,293.98

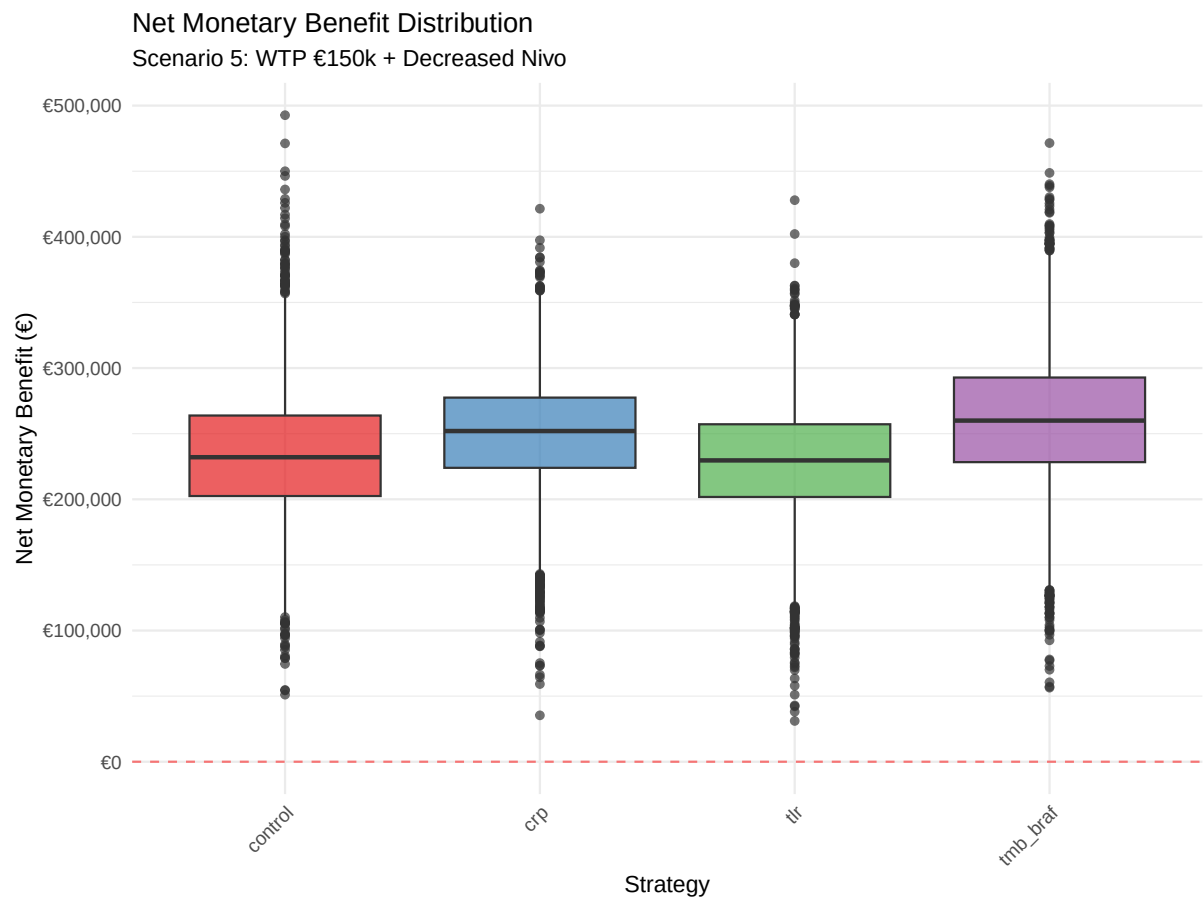


Figure 28: Net Monetary Benefit distribution (Scenario 5)

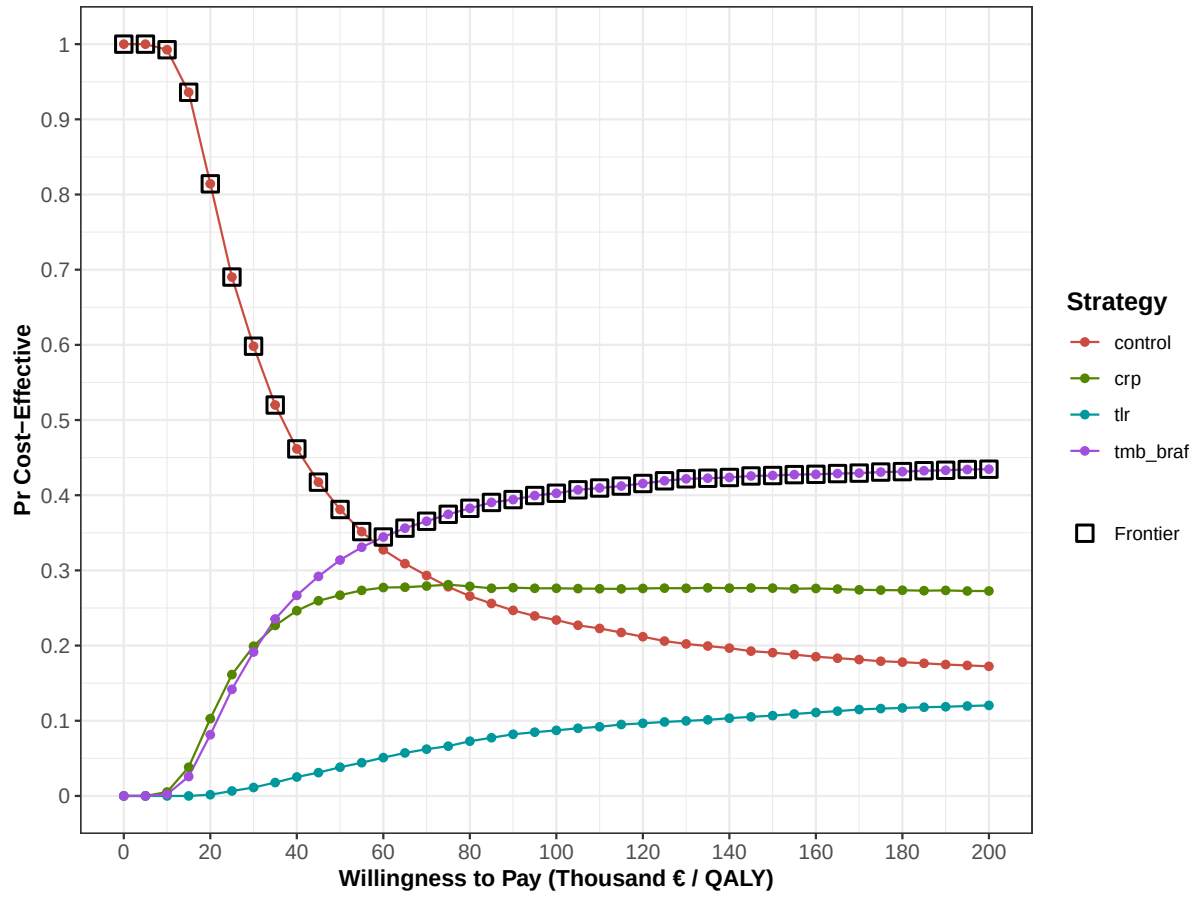


Figure 29: Cost-effectiveness acceptability curves (Scenario 5)

Total EVPI: $\text{€}2.301561 \times 10^4$

Table 13: EVPPI results: WTP €150,000 + decreased nivolumab

Parameter	EVPPI	% of EVPI
u_np	€3908.53	17%
u_p	€2514.18	10.9%
c_drug_FLOX	€458.93	2%
c_other_follow	€70.06	0.3%
c_other_visit	€61.29	0.3%
c_test_CT	€59.75	0.3%
c_other_last	€14.69	0.1%
c_drug_nivo	€0	0%
c_test_NGS	€0	0%
c_other_baseline	€0	0%

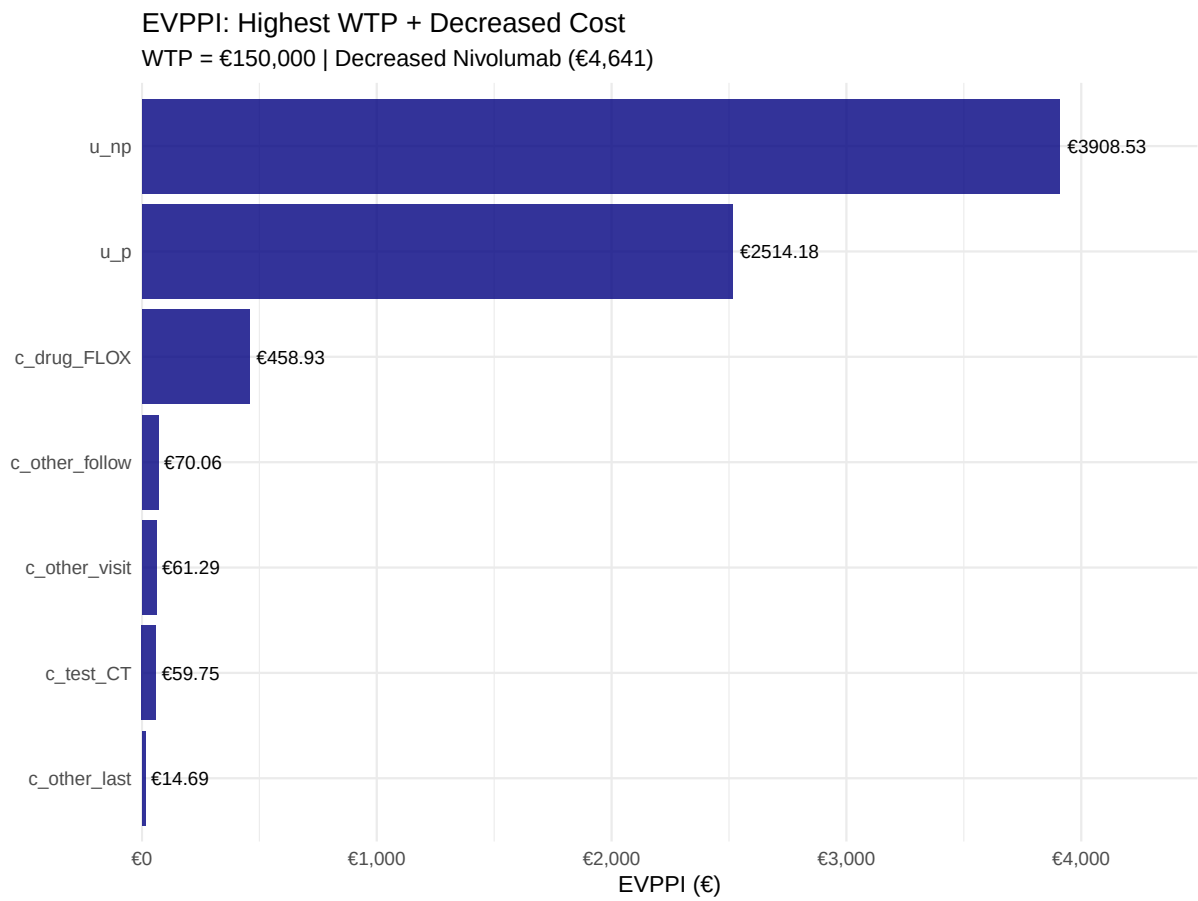


Figure 30: EVPPI: Highest WTP with decreased nivolumab cost

8 Comparative Analysis Across Scenarios

8.1 EVPI

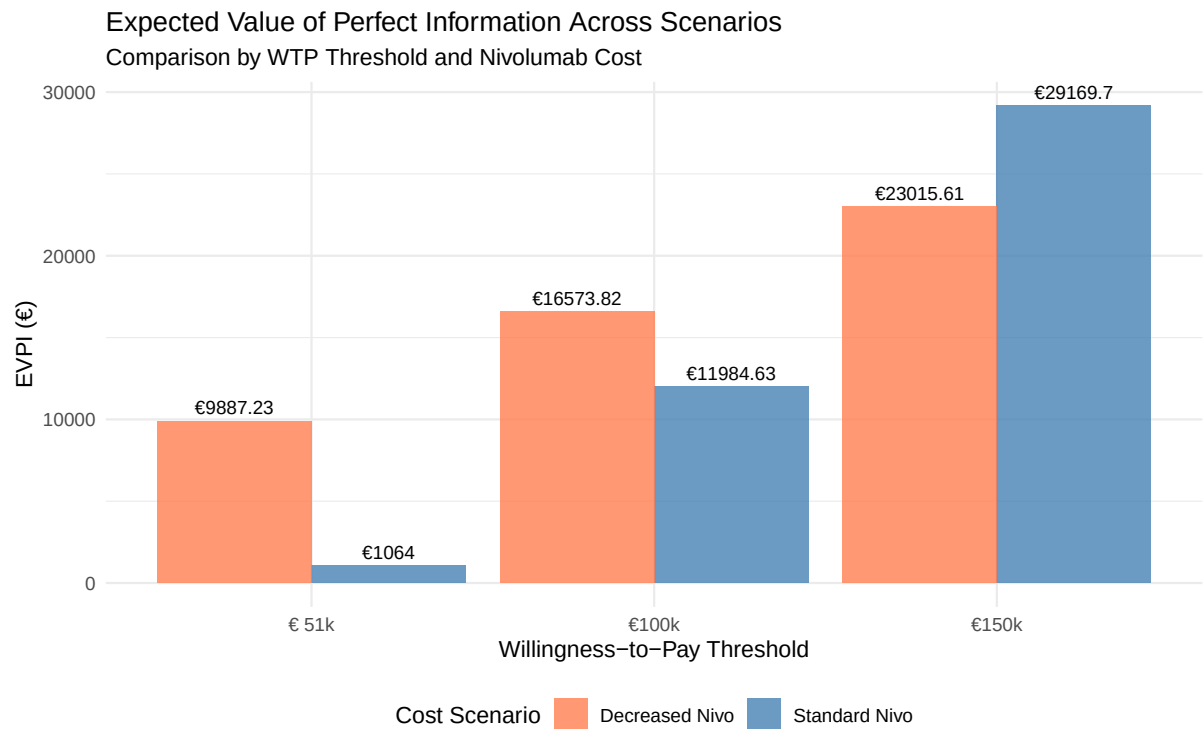


Figure 31: Expected Value of Perfect Information across scenarios

8.2 EVPPI

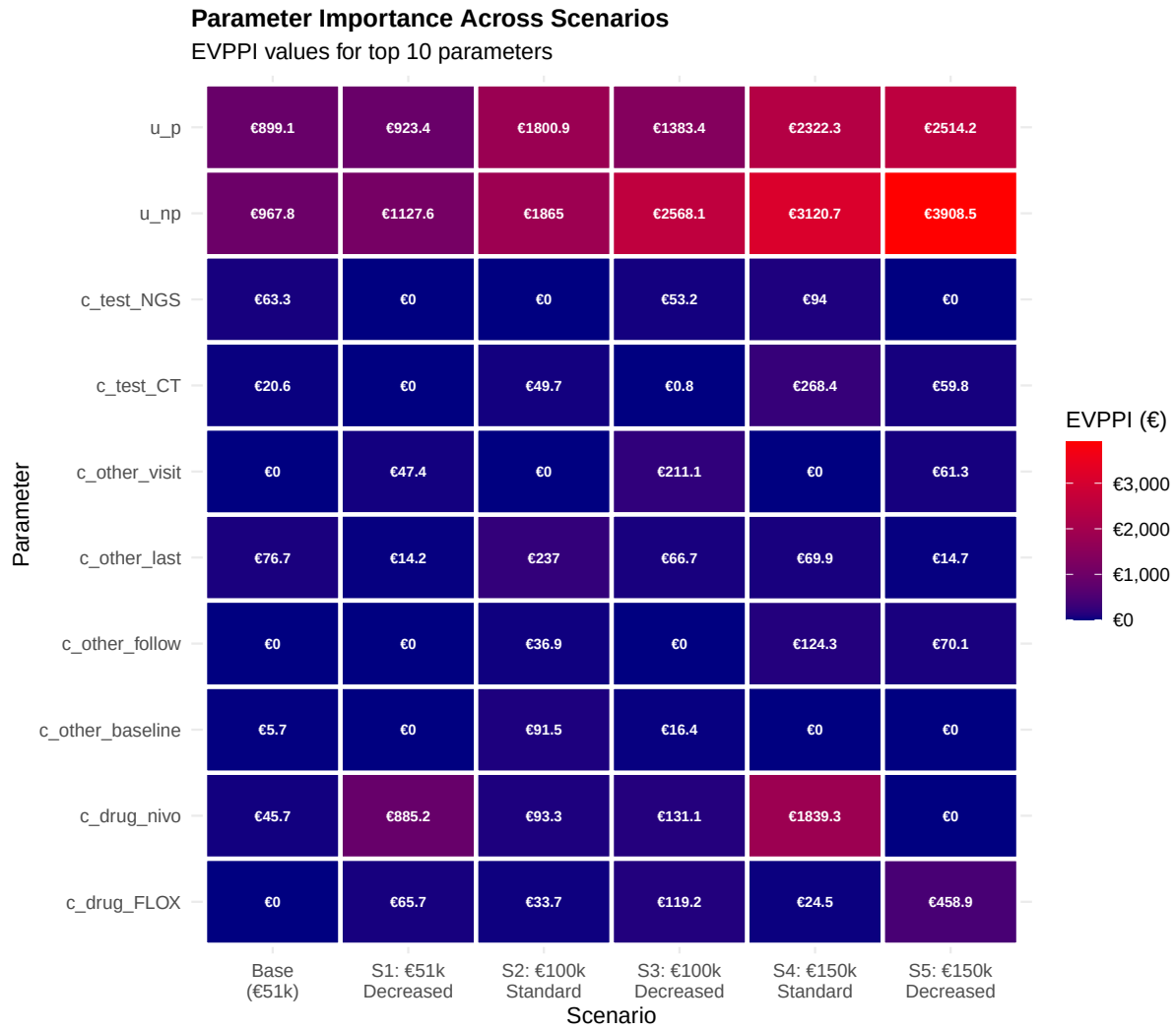


Figure 32: Parameter importance across scenarios (top 10 parameters)

8.3 Summary

Table 14: Summary of EVPI and top EVPPI by scenario

Scenario	WTP	EVPI	Top Parameter	Max EVPPI	% of EVPI
Base Case	€ 51,000	€1064	u_np	€967.84	91.0
Decreased Nivolumab Cost	€ 51,000	€9887.23	u_np	€1127.63	11.4
WTP €100,000	€100,000	€11984.63	u_np	€1865.01	15.6
WTP €100,000 + Decreased Nivolumab	€100,000	€16573.82	u_np	€2568.1	15.5
WTP €150,000	€150,000	€29169.7	u_np	€3120.65	10.7
WTP €150,000 + Decreased Nivolumab	€150,000	€23015.61	u_np	€3908.53	17.0

Scenario with Greatest Decision Uncertainty: WTP €150,000 (EVPI = €29169.7)

Most Important Parameter Across All Scenarios: u_np (Total EVPPI = €13557.76)

Impact of WTP Threshold: Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

Impact of Nivolumab Cost: Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.

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